

CLEAVAGES, POLITICAL PARTIES AND POLARIZATION

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Abstract

Why do some societies become more politically polarized and susceptible to populism? In this paper, I explore the hypothesis that these processes depend on how political parties respond to recent drivers within patterns of political competition that developed much earlier. Foundational political science theories argued that the dimensions along which political parties compete often originate in major conflicts that occurred in the period leading up to the emergence of mass politics. The societal divisions triggered by these conflicts became politically organized and shaped the creation of the modern party system, influencing political competition over the long run. I focus on the Carlist Wars (1833–1876) in Spain as an example of such conflicts and estimate their political effects using an instrumental variable exploiting variation in wartime logistical conditions combined with a novel dataset spanning three centuries to trace these effects over time. I find that areas more exposed to the wars exhibit lower political polarization today and weaker support for parties using populist rhetoric. Evidence on mechanisms suggests that the Carlist Wars helped form an additional axis of political competition that, insofar as it does not overlap strongly with the left-right divide, reduces incentives to intensify political polarization and weakens the conditions populist leaders often exploit. The findings remain after including a rich set of controls, and are not explained by the leading alternative hypotheses. Violations of the exclusion restriction would need to account for around 60% to 90% of the reduced-form effects to explain away the results, while placebo tests provide no evidence that the empirical strategy captures confounding factors. The results shed light on the role of historically rooted structures of political competition in shaping societies' susceptibility to polarization and populism.

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1. Introduction

Why do some societies become politically polarized and receptive to populist appeals, while others remain comparatively resistant? Existing research has focused primarily on relatively immediate causes, including immigration, technological, and economic shocks (Guriev & Papaioannou, 2022). However, the political system’s response to these shocks may vary depending on structural characteristics that were shaped much earlier, and we know little about their role in these processes.

In this paper, I focus on one characteristic and study whether the dimensions along which parties compete shape their incentives to intensify or moderate political polarization. Lipset and Rokkan (1967) famously argued that some of the most important dimensions of party competition can be traced to historical conflicts from the period leading up to the expansion of mass politics. According to this framework, disagreements arising from these conflicts created divisions, or “cleavages,” that politically mobilized people and shaped the formation of the modern party system. Once incorporated into the party system, these disagreements became institutionalized, continuing to mobilize voters and shape the dimensions along which parties compete over the long term.

These patterns of competition may help explain differences across societies in current trends of political polarization and populism. When political divisions “cross-cut” one another, the position of a voter on one dimension does not strongly predict their position on another. This would be the case, for instance, if knowing that a voter supports greater regional autonomy for parts of a country tells us little about whether they prefer left- or right-wing economic policies. In such cases, parties competing on both dimensions have less incentive to intensify polarization by taking extreme positions on one dimension because they may lose voters on the other. This can also weaken incentives to frame politics as a conflict between a unified “us” and “them” (Lipset, 1960; Goodin, 1975; Mason, 2016, 2018; Mason & Wronski, 2018; Levendusky, 2018; Harteveld, 2021). The opposite occurs when most major political divisions “reinforce” one another, so that the same people tend to cluster together across different dimensions². Therefore, historical conflicts that fostered cross-cutting divisions within the party system may make societies less vulnerable to polarization and populist mobilization over the long run. In this paper, I focus on the Carlist Wars in Spain as one such conflict and find evidence consistent with this mechanism.

² This would be the case if, for example, parties competed only along conservative–liberal and left–right economic dimensions, and most people who favored conservative policies also favored right-wing economic policies. In this case, knowing what someone supports on one dimension would be informative about what they support on the other.

The Carlist Wars were a series of civil wars arising from the 1833 succession crisis, after reforms allowed a woman, Isabella II, to inherit the throne. Her uncle Carlos challenged her succession, beginning a series of armed conflicts fought intermittently between 1833 and 1876 that initially affected much of Spain. As I discuss in Section 2.2, in areas most affected by the wars, the Carlists—who relied heavily on volunteers—increasingly sought support by politicizing potential grievances against the Crown. These included demands for the restoration of the *fueros*, regional legal privileges that granted greater control over taxation and administration and that, in many cases, had been abolished more than a century earlier. By making these aspects politically relevant and mobilizing people around them, the conflict would have helped create an additional dimension of political disagreement that was later incorporated into the party system through competition over regional autonomy and political centralization.

Consequently, the Carlist Wars are particularly well suited to this study because they occurred in the period leading up to the expansion of mass suffrage and helped politicize dimensions that later shaped political competition through the formation of political parties, in line with the historical processes described by Lipset and Rokkan (1967). The case also offers two empirical advantages. First, because the conflict was triggered by a dynastic succession crisis rather than by local or regional grievances, exposure to the war was not necessarily tied to places that were already more likely to mobilize around those issues, making it possible to study how war exposure shaped their political organization. Second, differences in the intensity of the wars across Spain allow me to examine how differential exposure shaped long-run political organization, polarization, and populist support across the country while holding national factors constant.

Empirically, I estimate the long-run effects of the Carlist Wars on political polarization and support for populism by exploiting local variation in exposure to battles³. A central identification concern is that battles may have occurred in places where political dynamics would have evolved as they did even in the absence of the war. For instance, battle locations may plausibly reflect areas where support for either side was initially stronger. To address this concern, I exploit historical evidence showing that the conflict persisted not only where Carlism initially enjoyed stronger support, but where logistical conditions limited the Crown’s ability to suppress the uprisings. These conditions would have reduced

³ Sections 2 and 3 discuss why this metric is informative in this setting.

the military imbalance between the Crown and the weaker Carlist forces, making resistance more viable and allowing the conflict to last longer in certain areas.

I leverage a configuration of geographic and infrastructural conditions that historical accounts identify as conducive to Carlist resistance. Specifically, I construct an instrument based on the interaction among distance to the Crown’s naval arsenals, mule-road density, and terrain ruggedness. This interaction captures their documented complementarity during the Carlist Wars: in rugged areas distant from Crown supply sites, mule roads enabled Carlist forces to engage in guerrilla warfare while limiting the Crown’s ability to deploy its military superiority effectively. The instrument therefore captures settings in which the Crown’s advantage was especially likely to break down, allowing resistance to persist. I isolate this war-related mechanism from potentially correlated societal characteristics by controlling for the independent and pairwise effects of the instrument’s components, as well as for a rich set of pre-treatment controls.

The exclusion restriction requires that, conditional on the lower-order terms and controls such as roads suitable for wheeled vehicles, the societal effects of proximity to the arsenals in rugged terrain do not vary systematically with mule-road density in the absence of the war. Put differently, among locations at the same distance from these supply sites and with similar terrain ruggedness, having more mule roads should not make a location systematically more affected by proximity to the arsenals once we condition on roads suitable for wheeled vehicles. This is plausible because the arsenals were part of broader Crown modernization efforts undertaken in the century preceding the war, which included investment in roads suitable for wheeled vehicles. Much of the mule-road network, by contrast, had developed centuries earlier for different purposes and was poorly suited to transporting artillery, machinery, and other heavy inputs used in military modernization. Once roads suitable for wheeled vehicles are taken into account, mule roads therefore arguably provided little additional peacetime-relevant connectivity, especially in rugged areas distant from the arsenals. Nonetheless, they became militarily valuable during the wars by facilitating Carlist guerrilla resistance. Consistent with this interpretation, the instrument is uncorrelated with a rich set of pre-war characteristics, does not predict exposure to conflicts lacking the same military asymmetry, and does not predict polarization in neighboring Portugal.

I find that greater exposure to battles during the Carlist Wars is associated with persistently lower political polarization and weaker support for parties using populist rhetoric in post-2010 elections. While the paper’s primary contribution is to identify a broader mechanism rather than the precise magnitude

of these long-run effects, the estimates nevertheless provide a useful sense of scale. Relative to the rest of Spain, average exposure among exposed locations of around four battles is associated with a 9.6-point decline on the 0–100 RQ polarization index and declines of 0.752 standard deviations on the Dalton polarization index and 0.636 standard deviations in reliance on populist rhetoric.

Because the exclusion restriction cannot be empirically verified, I assess how large a violation would need to be to overturn the paper’s directional conclusion. Sensitivity analyses following Conley, Hansen, and Rossi (2012) indicate that a direct effect would need to account for 61% to 91% of the reduced-form effect to explain away the results. The balance tests, placebo exercises, and evidence against the main alternative explanations make a violation of this magnitude unlikely.

To trace how these patterns emerged, I estimate the impact of wartime exposure on intermediate outcomes in the decades following the wars and use mediation analyses to connect these processes over time. Both the IV estimates on intermediate outcomes and the mediation analyses suggest that the effects are most plausibly explained by the increased salience of a cross-cutting center-region⁴ cleavage and greater support for regional and local political projects. Results obtained using Kwon and Roth’s (2026) methodology do not suggest that an alternative channel mediates the relationship between the Carlist Wars and contemporary patterns. Moreover, alternative explanations that could act as confounders also do not appear to mediate this relationship. The main results also remain robust when accounting for historical polities and linguistic boundaries, suggesting that, while these aspects play an important role in Spain’s center-region competition, what seems to matter is not whether those dimensions exist but how they became organized and represented politically through events such as the Carlist Wars.

This paper makes several contributions. First, it offers an alternative approach to understanding polarization and populist support. It complements the literature emphasizing the role of shorter-term cultural, economic, technological, and immigration shocks (Frey et al., 2018; Norris & Inglehart, 2019; Dustmann et al., 2019; Autor et al., 2020; Acemoglu & Restrepo, 2020; Broz et al., 2021; Guriev & Papaioannou, 2022; Dal Bó et al., 2023) by providing evidence that historically determined structures of political competition may help explain why some societies are more vulnerable to polarization and populist appeals than others. Second, it contributes to quantitative historical political economy in

⁴ Also known in the literature as the center-periphery dimension. Cross-cuttingness in this context decreases with the extent to which support for peripheral demands predicts an individual’s position on the left-right axis, based on voting patterns and party ideology along both dimensions.

economics, which has studied the long-run effects of historical shocks on attitudes, cultural traits, and institutions⁵. I build on this approach by examining how such shocks may also shape political cleavages and party competition. Third, it contributes to research on the consequences of civil war and hostile environments for long-term cohesion, political participation, and collective action (Blattman, 2009; Bellows & Miguel, 2009; Voors et al., 2012; Gilligan, Pasquale, & Samii, 2014; Lowes & Montero, 2021a; Tur-Prats & Valencia Caicedo, 2025). I extend this literature by examining additional long-run outcomes, including political polarization and support for populism. Finally, the paper contributes to Spanish economic history by studying the consequences of the Carlist Wars for contemporary political outcomes and by digitizing and combining a large body of historical material spanning three centuries.

The remainder of the paper proceeds as follows. Section 2 provides the contextual and historical foundations of the study. Section 3 develops the theoretical framework, and Section 4 describes the data and empirical strategy. Section 5 presents the main results. Sections 6 and 7 assess the validity of the empirical framework through placebo tests and robustness checks. Section 8 explores mechanisms of persistence, and Section 9 evaluates competing hypotheses and mechanism attribution. Section 10 discusses the study’s limitations, external validity, and implications for future research. Section 11 concludes.

2. Historical foundations - The Carlist Wars and the structure of political competition

Lipset and Rokkan (1967) argued that European party systems were largely shaped by divisions triggered by major historical conflicts in the period leading up to the expansion of mass suffrage. The transformations that accompanied the transition from traditional to modern societies involved political and economic reconfigurations that created winners and losers, around which people became politically organized. For instance, the Industrial Revolution generated new conflicts structured around class. The development of the modern nation-state⁶, including its control over education, marriage, and law, also created conflict with religious authorities that had previously exercised those functions. As mass politics expanded, these divisions were incorporated into the political system through the development of the first political parties. Once institutionalized, parties continued to mobilize voters and compete for

⁵ This literature is extensive. Some examples include Engerman and Sokoloff (1997), Acemoglu, Johnson, and Robinson (2001), Dell (2010), Nunn and Wantchekon (2011), Voigtländer and Voth (2012), Bleakley and Lin (2012), Alesina, Giuliano, and Nunn (2013), Nunn and Puga (2012), Dell, Lane, and Querubin (2018), Lowes and Montero (2021a, 2021b).

⁶ The modern nation-state is a sovereign political community united around the idea of a nation, rather than allegiance to a particular ruler. It exercises authority over a defined territory and exercises key functions such as education, law enforcement, and the military.

support around these divisions, preserving their relevance over time⁷. According to this framework, understanding which structural features shape contemporary political outcomes requires examining how these formative historical processes unfolded.

As I discuss in this section, the Carlist Wars are a natural setting in which to study these processes in Spain. They occurred in the period leading up to the expansion of suffrage, mass-party formation, and democratization, and were organized around grievances generated by Spain's process of nation-state formation⁸. This setting is also empirically useful because the spatial distribution of the conflict was shaped in important part by wartime logistics, which affected where the war could persist, rather than simply by where Carlism enjoyed stronger ideological support. I provide qualitative evidence that, in some of these places, the Carlists helped create a new dimension along which parties would later compete. Because the war was not triggered by these dimensions, and its spatial distribution was not determined solely by political factors, this setting makes it possible to study their effects on later political outcomes.

2.1. Why the Carlist Wars?

The Carlist Wars were a series of civil conflicts that occurred intermittently between 1833 and 1876, beginning with the succession crisis of 1833. Lacking a male heir, King Ferdinand VII enacted reforms that allowed his daughter, Isabella II, to inherit the throne. His brother, Carlos, contested the decision, arguing that succession should follow the traditional male line (Bullón de Mendoza, 1992). As a result, the coalition that formed around Isabella II and her mother, María Cristina, became dependent on liberal and reformist elites, whose support the regency needed in its struggle against Carlos (San Narciso et al., 2021). Carlos, by contrast, grounded his claim in the defense of tradition and drew support from sectors engaged in a counter-revolutionary effort to preserve or restore principles of the Old Regime, including Catholic monarchy and political absolutism (Rújula, 2023; San Narciso et al., 2021).

Consequently, the wars, which lasted intermittently for over forty years and with varying intensity across the country, came to represent conflicts around the grievances arising from Spain's transition from the Old Regime to the modern nation-state. They exposed the masses to the divisions created by these grievances, transforming them into dimensions around which different sides sought support and

⁷ Lipset and Rokkan (1967) argue that, by the early twentieth century, party systems had “frozen” around these dimensions.

⁸ For instance, Martin-Echebarria (2025) places the Carlist Wars within the broader wave of upheavals that followed the French Revolution, all part of the struggle over the transition from the Old Regime to liberal modernity.

along which people could organize and mobilize (Martin-Echebarria, 2025). Importantly, these processes unfolded in the period leading up to the enactment of male mass suffrage in 1890 and the emergence of the first political parties. In the next subsection, I discuss how the Carlists sought support in affected areas, the ideological evolution of the conflict, and the factors that shaped its geographical distribution.

2.2. Wartime logistics and the politicization of local societal divisions

This subsection emphasizes two aspects that are central to the study. First, the Carlist Wars exposed local populations to an ideological conflict around which people could be mobilized and whose geography was shaped not only by ideological dimensions such as initial support for Carlism, but also by the logistical conditions that made sustained military resistance possible. Second, primary sources and historical accounts suggest that in affected areas, the Carlists sought support by giving territorial differences against the centralized Crown new political significance.

Initially, the conflict extended across much of Spain, with particularly strong Carlist support and organization in areas such as Castilla la Vieja, Extremadura, and Andalucía (Bullón de Mendoza, 1992). In many of these areas, however, the Crown's superior military capacity allowed it to suppress Carlist mobilization relatively quickly (Bullón de Mendoza, 1992; Recio Cuesta, 2014). By contrast, Carlist forces were better able to resist in mountainous areas where rudimentary transport networks, including "*precarious mule-trains*," reduced the Crown's military advantage and made guerrilla warfare more viable (Lawrence, 2019). The campaigns of General Cabrera, which continued to trouble Isabelline forces until the end of the First Carlist War, illustrate this dynamic (Lawrence, 2019).

Over time, Carlists became associated with regional interests such as the defense of the *fueros*⁹ in places like Navarre, where they were able to sustain effective resistance (Bullón de Mendoza, 1992). Yet the movement did not begin in Navarre, nor did it initially incorporate this regional dimension (Bullón de Mendoza, 1992; Shih, 1998, p. 46). Contemporary observers instead suggest that initially such associations emerged partly from the geographical distribution of the war, rather than from the original Carlist project. Henningsen (1836), a foreign volunteer who served with Carlist forces, described the association between Carlism and the *fueros* as a "*journalistic misconception*" (p. 70). Later testimony by the Carlist soldier Juan Zaratiegui similarly rejected this association, stating that "*The war that*

⁹ Regional legal privileges granting greater control over taxation and administrative authority.

began in this country in 1833 had no other objective than the defense of the rights of then Don Carlos to the crown of Spain” (Zaratiegui, 1845, pp. xi–xii)¹⁰.

Indeed, local grievances such as those emerging from the *fueros* were not initially politically articulated before, or during the initial stages of, the Carlist Wars. Direct observers at the time reported that “*of those carrying arms, not one in twenty knows even the signification of the word Fueros, although it may be familiar to his ears*” (Henningsen, 1836, pp. 70–71). However, over time, Carlists increasingly started to mobilize people around this dimension. By making resistance to the centralized Crown visible and locally experienced, the wars contributed to the formation of “*novel identities*,” in part through the framing of Crown troops as “*foreign invaders*” (Shih, 1998, p. 46; Linz, 1973).

Carlism subsequently leveraged the newly politicized¹¹ *fueros* to seek support in affected areas. For instance, by 1872, Carlist discourse had incorporated a clearer regionalist dimension, promising to restore the *fueros* of the former Crown of Aragon more than 150 years after their abolition (Carlos VII, 1872). New political movements representing the divide between regional autonomy and centralization later emerged and evolved separately from Carlism. However, as Shih (1998) argues, these movements “*owed much to Carlism, insofar as the latter provided the element of resistance to the center*” (p. 47). In fact, some contemporary parties can be linked to this broader Carlist legacy without being direct continuations of Carlism itself. The Basque Nationalist Party, for example, was founded by Sabino Arana, who came from a Carlist family and incorporated themes such as the recovery of the *fueros* into late nineteenth-century politics (Shih, 1998).

The discussion above highlights three points. First, the geographical distribution of the war, and thus which areas became most affected, depended not only on political factors but also on logistical conditions that facilitated Carlist resistance. Second, Carlism contributed to the development of regionalist movements that emphasized greater regional autonomy, including through the defense of the *fueros*. Finally, the politicization of these dimensions by the Carlists is most plausibly understood as part of their efforts to seek support and mobilize people in affected areas, as illustrated by the marked shift in Carlist discourse on these issues over time.

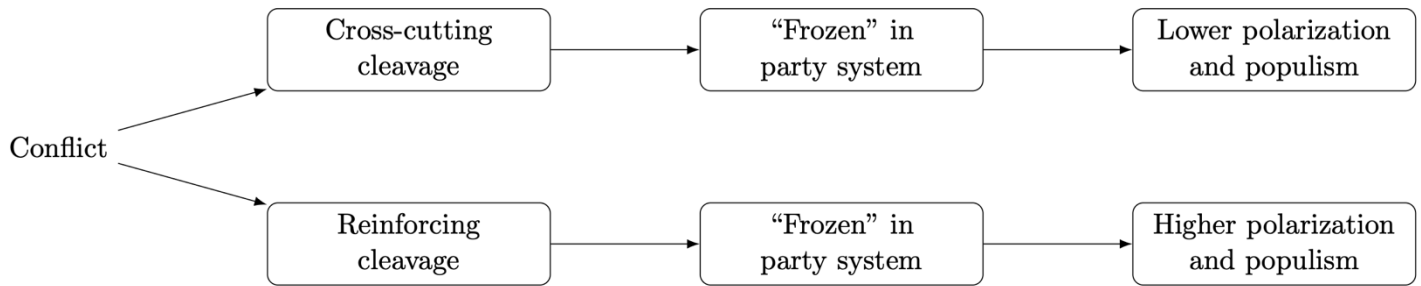
¹⁰ For a detailed contemporary account of the motivations behind the war and why soldiers chose to fight on the Carlist side, see Henningsen (1836).

¹¹ Politicized here means that it became a dimension along which people organized politically.

3. Theoretical framework

This section develops the theoretical link between the Carlist Wars, contemporary political polarization, and populist support. The argument can be summarized as follows: conflicts during the period leading up to the expansion of mass politics shape which political disagreements become institutionalized through the party system and, in turn, the dimensions along which parties compete over the long run. When these disagreements cross-cut one another—meaning that voters’ preferences on one dimension are not very informative about their preferences on another—parties have weaker incentives to move toward ideological extremes on any single axis because they compete across both dimensions and risk losing voters on the other. This constraint is weaker when the same voters consistently fall on the same side of a relevant political disagreement. Under these conditions, society is also easier to divide into two opposing camps, creating conditions that populist actors often exploit. Figure 1 summarizes this argument, while the rest of the section introduces the relevant concepts and explains each step in detail.

Figure 1: Theoretical diagram.



3.1. War and frozen political cleavages

Disagreements create social divisions, but not all of them are political cleavages. A political cleavage is a societal division organized around a disagreement that mobilizes citizens politically and along which parties can later compete for voters (Lipset & Rokkan, 1967).

Lipset and Rokkan (1967) argued that the major historical conflicts associated with nation-building and industrialization in the periods leading up to the expansion of mass democracy created political cleavages by mobilizing people around the grievances and disagreements generated by these processes. These disagreements were later translated into the political system through the formation of the first mass party systems, which “froze” these dimensions and shaped how parties compete and mobilize voters long after the conflicts that produced them had faded (Lipset & Rokkan, 1967; Bornschier, 2009).

While war is not necessary for this theory, it may make these divisions more durable by facilitating the creation of political and institutional structures (Tilly, 1992; North, Wallis, & Weingast, 2009) and shaping the transmission of political identities through schooling or other forms of socialization. For instance, Paglayan (2022) argues that the expansion of education after civil wars can serve as a tool of political indoctrination.

War also reshapes how people organize across social groups in affected areas. Although existing evidence is mixed on the direction of these effects, it consistently shows that war has important consequences for collective organization and political behavior. Some studies find that war and hostile environments weaken interpersonal trust among groups and individuals (Alesina & La Ferrara, 2002; Dyregrov et al., 2002; Tur-Prats & Valencia Caicedo, 2025). Others show that exposure to violence can, in some contexts, increase political participation, collective action, resource sharing, and local cohesion (Blattman, 2009; Bellows & Miguel, 2009; Voors et al., 2012; Gilligan, Pasquale, & Samii, 2014; Lowes & Montero, 2021a)¹².

Thus, wars triggered by the conflicts emphasized by Lipset and Rokkan (1967) should be expected to have marked effects on political organization around these disagreements and their institutionalization. Moreover, in this context, warfare is particularly relevant because, as discussed in Section 2, political organization along new dimensions was a way through which Carlists sought support for their volunteer base.

3.2. From cleavages and parties to political polarization and populism

I focus on one mechanism through which these processes may shape long-term patterns of polarization and populism: whether, at the time of the expansion of mass suffrage, parties institutionalized cross-cutting or reinforcing cleavages. Cross-cutting cleavages occur when citizens who share one political identity differ on another (Taylor & Rae, 1969). Reinforcing cleavages, by contrast, occur when political identities overlap, so that voters who align on one dimension also tend to align on others (Taylor & Rae, 1969; Goodin, 1975). This distinction matters because it shapes how parties compete, tending to either moderate or intensify political polarization over time.

¹² These counterintuitive effects may reflect changes in preferences and beliefs consistent with theories of post-traumatic growth, or shifts in local institutions and social norms that strengthen incentives for collective action (Tedeschi & Calhoun, 1996; Powell et al., 2003; Bellows & Miguel, 2009; Tilly, 1975).

Although political polarization is difficult to operationalize, Fiorina and Abrams (2008) argue that one of its most common definitions refers to a distribution of political preferences that becomes bimodal and concentrated around ideological extremes. Cross-cutting cleavages are expected to moderate polarization because they make it harder to divide society into two clearly defined opposing camps. They also reduce parties' incentives to adopt extreme positions on any single axis of competition, since doing so may repel voters who support the party on another dimension (Goodin, 1975; Dunning & Harrison, 2010). For instance, if regionalists on the center-region axis are equally likely to be left- or right-wing, a regionalist party has little incentive to move far to the right, as doing so could alienate its left-wing regionalist supporters¹³. Likewise, such a party would be unlikely to portray "leftists" as a unified adversarial group when some are part of its potential electoral base precisely because the cleavage is cross-cutting.

By contrast, when cleavages reinforce one another, movement toward ideological extremes on one dimension becomes less costly, since voters are likely to be aligned on other dimensions as well. For instance, if supporters of right-wing economic policies are also more likely to be socially conservative, a conservative party may move far to the economic right in response to a shock because conservative voters are also more likely to support such policies. Consequently, reinforcing cleavages are expected to intensify political polarization (Mason, 2016; Mason & Wronski, 2018; Harteveld, 2021).

Populism, in turn, is a political approach that frames politics as a moral struggle between a virtuous and unified "people" and a corrupt elite that has captured the political system (Mudde, 2004; Mudde & Rovira Kaltwasser, 2017). Although the literature has not converged on a single definition, populism is often also anti-pluralist and anti-institutional (Guriev & Papaioannou, 2022). By claiming to represent the "real people," populist leaders portray opponents or outsiders as illegitimate rather than as ordinary political competitors, and they justify bypassing democratic institutions that protect political pluralism (Müller, 2016; Guriev & Papaioannou, 2022). In this sense, populism provides a justificatory language for attacks on courts, legislatures, media, opposition parties, and electoral constraints, presenting them as necessary steps to restore the will of the people (Ruth, 2018).

Reinforcing cleavages and polarization can provide fertile ground for this populist logic. Because they make it easier to define a homogeneous "people" and to frame politics in "us versus them" terms, they

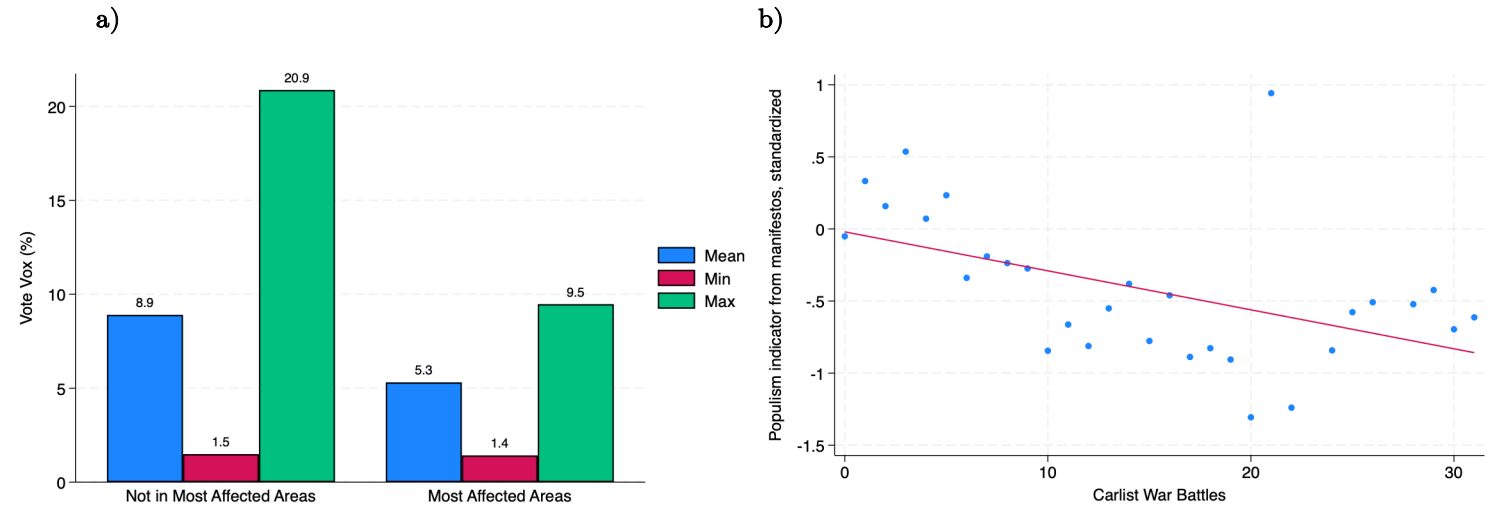
¹³ In the case where there are regionalist/left and regionalist/right parties, the same mechanism operates because parties must balance two dimensions. Going too far to the right or left may cause them to lose support on the center-periphery dimension because the two dimensions are cross-cutting.

facilitate the division of society into two morally opposed camps (McCoy, Rahman, & Somer, 2018). Polarization may also enable the anti-pluralist and anti-institutional components of populism. In highly polarized systems, evidence suggests that as the perceived cost of the opposing side winning increases, citizens become more willing to tolerate anti-democratic behavior by their own side (Svolik, 2019). They may even prefer co-partisan candidates who violate democratic principles over opposing-party candidates who respect them (Graham & Svolik, 2020). In this sense, polarization can create favorable conditions for populist projects to take hold. At the same time, populism may further polarize voters over time (Frantz et al., 2025), creating a feedback loop that may be difficult to break.

Illustration of this argument in the Spanish case: Figure 2 presents two descriptive patterns consistent with the argument that historical conflicts such as the Carlist Wars can shape political competition over time and, in turn, influence contemporary populism. Figure 2a shows that areas more exposed to the Carlist Wars display substantially lower contemporary support for Vox, Spain’s principal populist party according to [PopuList](#): its vote share is about 60 percent of that observed elsewhere. Since Vox differs from other parties along dimensions beyond populism, I also construct a party-level measure of anti-institutional and exclusionary populism using post-2010 manifestos from the Regional Manifestos Project¹⁴. Because each quasi-sentence is assigned to a single category, this measure helps distinguish populist rhetoric from broader ideological and territorial differences. Figure 2b reveals a similar relationship: greater battle exposure is associated with lower average levels of populism.

¹⁴ This indicator is described in more detail in Section 4.1 and Annex 2.

Figure 2: Carlist Wars and support for populist parties.



Notes: Panel a) shows Vox’s vote share in each of today’s autonomous communities defined by Tejerizo & Quirós (2023) as most affected by the Carlist Wars. “Min” represents the vote share in the region with the lowest support, and “Max” represents the vote share in the region with the highest support. “Mean” is the mean of the vote share among autonomous communities within each group. Panel b) shows a scatterplot of the average populist indicator by the number of Carlist Wars battles within a 25 km radius of each municipality. The populist indicator is created using principal component analysis, and is described in the rest of the paper.

4. Methodology and data

4.1. Dependent variables

I combine municipal-level electoral results with party-level ideological classifications from the Regional Manifestos Project (RMP) to construct measures of polarization and populism. These sources provide each party’s vote share in each municipality-year and its positions on selected ideological dimensions based on party manifestos from autonomous-community elections. For each RMP dimension, I use the average over the study period to classify political parties, as this provides a more stable ideological position that is less driven by election-specific considerations. One advantage of the RMP is that it classifies manifestos from regional parties and assigns region-specific ideological positions to national parties, better capturing geographic heterogeneity in political competition across Spain.

I use municipal elections as the baseline for two reasons. First, they are held simultaneously across Spain, reducing the risk that electoral outcomes are affected by national shocks occurring at different times. By contrast, elections in autonomous communities are held in different years. Second, municipal elections capture political competition and its effects at the local level while also being informative of cross-sectional ideological variation¹⁵ (Tausanovitch & Warshaw, 2014). Voting shares in general

¹⁵ People vote in these elections based on their ideological preferences too.

elections in Spain are more likely to reflect coalition-building dynamics, and elections may be repeated when coalition negotiations deadlock, as in 2015 and 2019. While parties in these elections are also likely to tailor their competition to local characteristics through targeted pamphlets, rallies, and other strategies, the resulting vote shares may be less representative of competition driven primarily by the local cleavage structure. Nonetheless, I also replicate the analysis using general-election results for robustness.

4.1.1. Political polarization

As discussed in Section 3.2, the most common definition of political polarization involves both the concentration of preferences at ideological extremes and bimodality (Fiorina & Abrams, 2008). I therefore combine two metrics to measure this process. To capture the distribution of preferences toward ideological extremes, I follow Dalton (2008), who proposes the indicator shown in Equation [1]. This indicator is essentially an ideological standard deviation weighted by vote shares.

Because a distribution may be dispersed without necessarily being bimodal, I complement this indicator with the RQ Polarization Index proposed by Montalvo and Reynal-Querol (2005), shown in Equation [2]. This index reaches its highest values when the distribution is concentrated around two relatively large groups. However, it does not capture ideological dispersion. I therefore use the two indicators as separately informative measures of different dimensions of polarization. A politically polarized society would score highly on both, reflecting both ideological extremism and bimodality.

$$[1] D_j = \sqrt{\sum_{i=1}^{N_j} (L_{ij} - \bar{L}_j)^2 \cdot v_{ij}}$$

$$[2] P_j = 4 \sum_{i=1}^{N_j} v_{ij}^2 (1 - v_{ij})$$

D_j denotes the Dalton Index and P_j the RQ Polarization Index in location j . L_{ij} denotes the ideological position of party i in location j , $\bar{L}_j$ the average ideological position in location j , v_{ij} the vote share of party i in that location, and N_j the number of parties competing in location j .

4.1.2. Support for populist agendas

Measuring populism involves several complications. First, beyond the two core features most commonly associated with populism—anti-elitism and anti-pluralism (Mudde, 2004; Müller, 2016; Guriev &

Papaioannou, 2022)—there is no consensus on where the concept should stop, or on which additional elements should be treated as constitutive of populism rather than merely associated with it (Guriev & Papaioannou, 2022). Second, populism is not directly observed. For this reason, rather than treating any single manifesto category as a direct measure of populism, I rely on 16 variables from the RMP that capture rhetorical and ideological dimensions commonly linked to populist discourse, including anti-institutionalism, exclusionary appeals, strong authority, anti-immigration frames, anti-internationalism, protectionism, and references to political corruption (Mudde, 2004; Müller, 2016; Urbinati, 2019; Basile & Mazzoleni, 2020; Verzichelli, 2020; Guriev & Papaioannou, 2022).

The empirical strategy is therefore not to operationalize the most minimal definition of populism, or only its “two core features,” but to identify a latent process through the way populism expresses itself across these related dimensions. This leverages the fact that, while each dimension may appear separately outside populist discourse, populist discourse almost always includes them in combination. Anti-internationalism and protectionism illustrate this point: although neither is equivalent to populism, populism almost always expresses itself, at least in part, through these dimensions (Verzichelli, 2020).

Thus, the dimensions considered here are not treated as inherently populist. Rather, they are interpreted as indicators of populism when they co-vary with other rhetorical dimensions associated with it. If a latent component simultaneously increases across these dimensions, it is more likely to capture populism than an alternative ideological construct.

To measure this latent process, I group the 16 variables into three dimensions of populism and use principal component analysis (PCA) to estimate their structure across 121 parties in Spain. Annex 2 provides a detailed discussion of the construction of the indicator and the PCA results. These results suggest retaining only the first component, which explains around 51% of the variance among the included variables. The factor loadings indicate that this first component increases with anti-multiculturalism, opposition to equal rights for immigrants, other anti-immigration framings, anti-constitutionalism, strong authority, references to political corruption, anti-internationalism, and protectionism. I interpret the process that moves these variables in that direction as anti-institutional and exclusionary populist rhetoric¹⁶, consistent with “the main features of modern populism” identified by Guriev and Papaioannou (2022).

¹⁶ For simplicity, I refer to this indicator as “populism,” while acknowledging that it may not capture the concept’s most minimal definition.

I then use this indicator to calculate each location’s vote-share-weighted average level of populism, as shown in Equation [3].

$$[3] \text{ Populism}_j = \sum_{i=1}^{N_j} v_{ij} \text{PopulismFC}_{ij}$$

Here, PopulismFC_{ij} is the predicted first component score for party i in location j . v_{ij} is the vote share for party i in location j , and N_j denotes the number of parties competing in location j .

4.2. Identification strategy

Because the study asks a directional, mechanism-based question, it is not primarily concerned with estimating the precise magnitude of the effect of nineteenth-century wars on contemporary outcomes. The identification strategy must therefore address alternative explanations that could account for the direction of the estimated relationship or obscure the mechanisms behind it. In some settings, conditioning on variables that capture alternative channels of influence may be sufficient. In the present case, however, regressing political polarization and populism on war exposure, even conditional on pre-treatment observables, remains problematic for two related reasons.

The first concern is that battle locations may have been endogenous to pre-existing societal characteristics that could themselves explain contemporary political outcomes, even in the absence of conflict. This concern is plausible here since, as discussed in Section 2.2, battles may reflect the Crown’s efforts to suppress insurgency in areas with stronger Carlist support.

The second concern is that the long interval between nineteenth-century conflict and present-day outcomes creates scope for numerous intervening developments to confound the relationship. Areas more heavily affected by the war may have subsequently followed distinct historical trajectories for reasons unrelated to the conflict itself. In that case, the estimates may capture the effects of later developments—such as differences in migrant populations or other independent historical processes—rather than the long-run consequences of war exposure.

To address these concerns, I construct a historically informed instrumental variable based on the idea that some locations experienced more battles because logistical conditions limited the Crown’s ability to repress Carlist insurrections, rather than because of ideological or political factors. The next subsections develop this logic and explain why these conditions are unlikely, for military and

technological reasons, to have affected later political outcomes through channels unrelated to the Carlist Wars.

4.2.1. Estimating equations and instrumental variable

I use a two-stage IV specification—Equations [4] and [5]—to estimate the long-term effects of the Carlist Wars.

$$[4] \quad War_j = \alpha^f + \gamma^f X_j + \delta_1^f DCA_j \cdot Rug_j + \delta_2^f Mule_j \cdot Rug_j + \delta_3^f Mule_j \cdot DCA_j \\ + \omega^f \underbrace{DCA_j \cdot Rug_j \cdot Mule_j}_{\text{Excluded term}} + \varepsilon_j^f$$

$$[5] \quad Y_j = \alpha + \gamma X_j + \delta_1 DCA_j \cdot Rug_j + \delta_2 Mule_j \cdot Rug_j + \delta_3 Mule_j \cdot DCA_j + \beta \widehat{War}_j + \varepsilon_j$$

Here j represents a 10 km×10 km grid cell. Y_j corresponds to the political polarization and populism indicators discussed earlier. X_j is a vector of control variables including the levels of the components of the instrument and roads suitable for wheeled vehicles, DCA_j is the distance from the centroid to the nearest Crown naval arsenal, $Mule_j$ is the density of mule roads in the grid, and Rug_j is terrain ruggedness of the grid. War_j is the number of times the grid cell is within a 25 km radius of a Carlist Wars battle¹⁷. $DCA_j \cdot Rug_j \cdot Mule_j$ represents the instrumental variable.

Relevance: Building on the discussion in Section 2.2, the instrument is designed to predict locations where more battles would be fought because, in such places, the military asymmetry between the Crown and the Carlists should have been reduced, facilitating Carlist resistance. The instrument is modeled as an interaction because it captures how these conditions operated complementarily. Rugged terrain without mule roads constrained the mobility of both sides, while mule roads in flatter terrain were less effective at offsetting the Crown’s advantage in moving troops and supplies. Where mule roads crossed rugged terrain (Lawrence, 2019), however, Carlist forces could remain mobile and engage in guerrilla warfare under conditions that made it more difficult for the Crown to exploit its superior military resources. This reduction in asymmetry should have been further enhanced by distance from the Crown’s naval arsenals, given their important role for the Isabelline army during the wars (Fernández Gaytán, 1987; Pardo San Gil, 2006). Areas combining rugged terrain, dense mule-road networks, and

¹⁷ The annex reports results using a range of alternative radii. A 25 km radius is consistent with the spatial scale of some military operations documented in engagements such as the Action of Artaza, which unfolded across a valley system spanning roughly 20–30 km. For reference, 25 km corresponds approximately to a day’s travel on foot at the time.

greater distance should therefore have been more likely to experience sustained resistance and repeated armed encounters. This is also consistent with broader literature suggesting that terrain favorable to guerrilla warfare can restrict the effective reach of state military capacity and allow insurgencies to persist, especially in areas located far from state supply sites (Fearon & Laitin, 2003; Buhaug, Gates, & Lujala, 2009).

While other functional forms may also capture this complementarity, interactions are transparent and widely used in linear regressions. Moreover, there are no clear theoretical reasons to expect an alternative functional form to change the predicted direction of the relationship, which is the main empirical concern.

Exclusion: The instrument predicts specific locations that may differ in ways other than exposure to the war. For instance, these locations may have had lower Crown military access in peacetime, which could independently affect long-term political dynamics. The specification addresses these concerns by controlling for both the levels of the instrument components and their pairwise interactions. The levels capture the distinct historical trajectories associated with each characteristic, while the interactions allow their combinations to encode additional information when one characteristic may have shaped or conditioned the other.

In this interpretation, $Mule_j$ captures the distinct historical evolution of locations connected to pre-modern trade; DCA_j captures potential effects of proximity to major naval and military bases, including peacetime access; and Rug_j captures potential isolation and other characteristics that may have led societies in rugged areas to evolve differently¹⁸. The interaction $Mule_j \cdot Rug_j$ allows locations that were both connected to pre-modern trade and geographically rugged to follow a separate trajectory, as locally built mule roads were likely shaped by ruggedness and jointly reflect distinct societal characteristics. Similarly, $DCA_j \cdot Rug_j$ allows the influence of proximity to major naval and military bases to vary with terrain ruggedness, since rugged terrain may have constrained military operations from the arsenals by hindering artillery movement, while also reflecting differences in peacetime access. Because the proposed military complementarity is expected to arise primarily in rugged terrain, $Mule_j \cdot DCA_j$ is included as a lower-order control rather than as an excluded instrument¹⁹.

¹⁸ For instance, Nunn and Puga (2012) document differential effects of this variable in Africa and Asia.

¹⁹ The historical mechanism does not provide a clear reason to expect mule-road density and arsenal distance to jointly predict war exposure outside rugged terrain. Consistent with this interpretation, $Mule_j \cdot DCA_j$ has little independent first-

Thus, the specification clarifies the identification assumption required for the exclusion restriction to hold. Conditional on the lower-order terms and the remaining controls, including roads suitable for wheeled vehicles, using $DCA_j \cdot Rug_j \cdot Mule_j$ as the excluded instrument requires that, absent the conflict, the effect of $DCA_j \cdot Rug_j$ on later outcomes did not vary systematically with mule-road density. In words, conditional on roads suitable for wheeled vehicles, Crown peacetime military access from the arsenals in rugged terrain should not have been systematically facilitated by differences in mule-road density, nor should such differences have systematically influenced other historical processes unrelated to the conflict that affected later outcomes.

This assumption is historically plausible because mule roads and Crown naval arsenals developed through largely separate institutional processes and for purposes that imply very limited logistical complementarity in peacetime²⁰. Mule roads would have provided little additional access once roads suitable for wheeled vehicles are taken into account, especially in rugged areas distant from the arsenals. Crown naval arsenals were established in the eighteenth century as part of Bourbon modernization, alongside more modern road infrastructure, and were oriented toward imperial warfare and external defense (De Diego García, 2002; Kuethe & Andrien, 2014). By contrast, the main mule-road networks are documented centuries earlier and were intended primarily for travel on foot or horseback. They were poorly suited to the heavy artillery and wheeled transport associated with the arsenals' logistical and military functions²¹. Mule-road density and distance to the arsenals are therefore unlikely to have jointly reflected a distinct peacetime historical trajectory beyond their independent effects or, conditional on roads suitable for wheeled vehicles, to have facilitated Crown peacetime access from the arsenals. Moreover, any such complementarity became even less plausible after the Carlist Wars, as wheeled transport grew increasingly important. Therefore, if the three-way interaction has a large effect on outcomes, it is most likely capturing the wartime value of mule roads for sustaining Carlist resistance rather than a distinct post-war development channel.

stage predictive power once the remaining lower-order terms are included, and the estimates are similar when this interaction is omitted. Using a weak instrument as an excluded source of variation can bias IV estimates, whereas including the same plausibly exogenous variable as a control may improve precision.

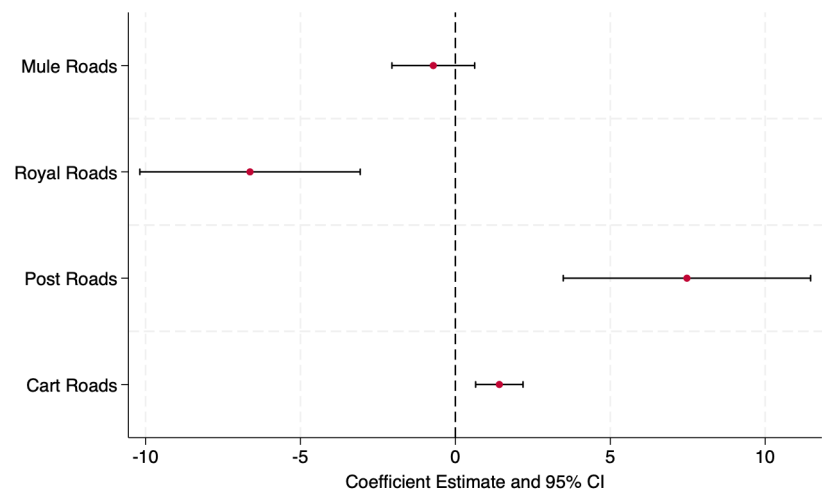
²⁰ Existing case studies and syntheses suggest substantial continuity between pre-1600 routes and many of the principal historic mule roads later recorded in documentary sources, including nineteenth-century maps (Diago & Ladero, 2009; Recondo, 2024; Pastor & García, 2006; Aguarod & Lostal, 1982). For context, the Bourbon dynasty came to power in Spain in 1700.

²¹ For Bourbon investment in paved roads, see Diago and Ladero (2009, especially section 7); on royal roads as the backbone of the Bourbon territorial plan, see Reguera Rodríguez (1999).

Correlational evidence supports this interpretation. Figure 3 shows that, unlike other road types, mule-road density is a poor predictor of distance to the nearest Crown naval arsenal. This suggests that the availability of mule roads did not systematically influence arsenal location and is therefore consistent with the notion that they were not the most relevant axis of connectivity to the arsenals. Thus, if two locations are at a similar distance from the arsenals, have similar ruggedness, and are also similar in other road infrastructure, differences in mule-road density are unlikely to explain why one location would be more affected by the arsenals than the other.

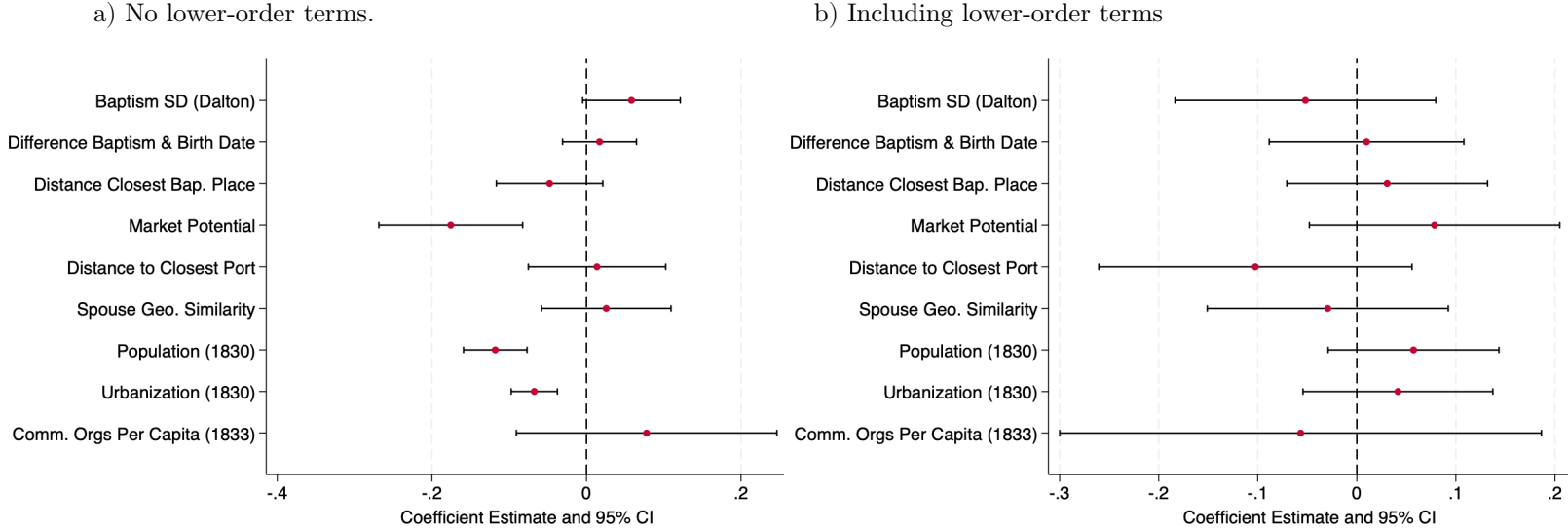
Moreover, if the instrument were capturing confounding channels other than wartime exposure, it would likely predict at least some important pre-treatment characteristics across the country. Figure 4 reports coefficients from regressions of each pre-war observable on the instrument. Figure 4a presents estimates without controls for the lower-order terms of the instrument’s constituent variables, while Figure 4b includes these additional controls. The instrument is negatively correlated with market potential, population, and urbanization; however, these correlations become statistically insignificant after including the lower-order terms. This pattern is consistent with the lower-order terms capturing the primary channels through which each component of the instrument, and their interrelationship, may affect broader societal and economic outcomes.

Figure 3: Distance to Crown naval arsenals and historical road density.



Notes: Figure 3 shows the coefficients of the effect of grid-level density of each road type on distance to the closest Crown naval arsenal. Standard errors are clustered at the NUTS-3 level.

Figure 4: Correlation between the instrumental variable and pre-war observables.



Notes: Pre-War Difference Baptism & Birth Date is the average number of days that elapsed between baptism date and birth date in each baptism place. Pre-War Baptism S.D. (Dalton) is the standard deviation of the days elapsed between baptism date and birth date in each baptism place. Pre-War Spouse Geo. Similarity is the average in each baptism location of the cosine similarity between the geographic distribution of the surnames of the father and the mother of each baptized child. Pre-War Market Potential uses the time of transportation as cost. Comm. Orgs Per Capita represents the number of commercial organizations divided by population. “Lower-order terms” refers to the levels and pairwise interactions of mule road density, terrain ruggedness and distance to the closest Crown naval arsenal. All variables are standardized. No additional controls are included in the specifications.

4.2.2. Control variables

While the historical argument provides a basis for the exclusion restriction, and correlational evidence supports these interpretations, the possibility that the identifying variation may be capturing residual channels of influence cannot be ruled out entirely. To further mitigate these concerns, I include a rich set of controls in X_j and assess the credibility of the identifying assumption through placebo tests, validity checks, sensitivity bounds, and an evaluation of leading alternative hypotheses.

The set of pre-war controls accounts for the average historical evolution predicted by observable pre-war characteristics, helping rule out a range of alternative mechanisms.

I begin with economic indicators: population, urbanization, the presence of roads suitable for wheeled vehicles, distance to the nearest main port²², and a measure of market potential (Harris, 1954). The

²² These ports were chosen based on historical evidence that they played a major role in the country’s connectivity and economic activity before 1833 (Rueda, Sazatornil, & Delgado, 2008). This selection criterion improves comparability by ensuring that treated locations are proximate to ports of similar historical importance, rather than mixing major ports with minor or rudimentary ones.

latter captures market access by measuring travel costs from each grid cell to major commercial and administrative hubs such as Madrid. These characteristics may have shaped the diversity and communal organization of local societies, as well as subsequent migration, industrialization, and the emergence of class cleavages. I also control for maximum caloric suitability and irrigation, which measure the potential for intensive agrarian activity—a relevant source of political organization given the emergence of several agrarian parties in Spain before 1936. Finally, commercial organizations per inhabitant serve as a proxy for both local economic organization and the broader capacity to organize collective interests. Together, these variables help address concerns that the estimated relationship is driven by differences in population, urbanization, commercial connectivity, organizational capacity, or agrarian potential.

I next include available societal indicators that capture dimensions not fully reflected in the economic controls. I proxy for religiosity using the interval between birth and baptism recorded in baptismal registers. Shorter intervals indicate closer adherence to Catholic prescriptions that children be baptized soon after birth to protect them from original sin in the event of early death²³. Consistent with this interpretation, Marco-Gracia (2022), for instance, finds that baptismal delays increase as child mortality declines. I calculate the average interval for each baptism site, as well as a Dalton polarization measure along the same dimension. The first measures average adherence among families baptizing children at the site, while the second measures heterogeneity in such adherence. Because the indicator may also reflect the cost of reaching a church, I control for distance to the nearest baptism site. These variables account for differences in both average religiosity and its local dispersion.

Finally, I construct a spousal geographic similarity indicator measuring the importance of forming families with partners of similar geographical origin²⁴. For each surname, I represent its pre-war geographical distribution as a vector and, for baptisms recorded up to 1833, calculate the cosine similarity between the vectors corresponding to the father’s and mother’s surnames²⁵. Higher values indicate that individuals more often formed families with partners whose surnames had similar

²³ See Annex 2 for a discussion of the construction of this indicator and the limitations of using baptism records to derive it.

²⁴ More generally, isonymy and related surname-based approaches use patterns of surname sharing and geographic distribution to estimate population relatedness, regional endogamy, inbreeding, and historical population structure (Crow & Mange, 1965; Lasker & Kaplan, 1985; Jobling, 2001; Mateos, 2007), and have been applied in the Spanish context (Rodríguez-Larralde et al., 2003).

²⁵ Under Spanish naming conventions, people typically have two surnames: the paternal surname followed by the maternal surname, e.g., Pedro López Pérez.

geographical origins. The measure therefore accounts for differences in the preservation of geographical lineage and the local social structures associated with it.

4.3. Data

I construct a novel dataset that combines historical and contemporary sources at the $10 \text{ km} \times 10 \text{ km}$ grid-cell level. Because Spanish administrative boundaries changed over time, I use grid cells as the unit of analysis and harmonize variables observed at different spatial scales using geographic overlap. This is the finest resolution at which the main controls can be consistently incorporated.

The dependent variables are constructed by combining data from Spanish municipal elections since 2011 with the RMP data²⁶ described in Sections 4.1.1 and 4.1.2. The analysis begins in 2011 because it coincides with the period in which contemporary democratic backsliding intensified and became a central focus of comparative political research (Bermeo, 2016; Lührmann & Lindberg, 2019; Maerz et al., 2020; Hellmeier et al., 2021). The main historical treatment is exposure to the Carlist Wars. I compile documented battles from historical and academic sources, starting from Wikidata entries and cross-checking the cited references. I also add battles absent from Wikidata when documented in the consulted sources. For each event, I record its location, year, and available characteristics, and georeference locations using OpenStreetMap, the ChatGPT API, and manual verification. I apply the same procedure to events from the Peninsular War and Spanish Civil War for placebo exercises.

To construct the instrumental variable, I combine geographic and historical transport data: the Terrain Ruggedness Index from Amatulli et al. (2018), manually digitized historical roads from the *Mapa general de Caminos de España y de Portugal*, and Crown naval arsenal locations geocoded using OpenStreetMap.

I use 1830 HYDE data to measure pre-war urbanization and irrigation. I estimate 1830 population by applying HYDE-implied growth rates to the 1787 census population. Additional controls include maximum potential caloric output from Galor and Özak (2016), market potential based on digitized roads and historical population following Harris (1954), and commercial organizations per capita constructed from digitized records of the *Censo Electoral de Asociaciones Profesionales y Relación de las Instituciones no Profesionales de Ahorro, Cooperación y Previsión*.

²⁶ On average across municipalities, approximately 87% of votes in municipal elections correspond to parties with a matching entry in the RMP. Indicator weights based on the RMP are computed using only votes cast for those parties.

For analyses including Portugal, I apply the same procedures whenever homologous variables are available. Because the RMP does not cover Portugal, I use the RQ index for Portugal placebo tests. Most variables have Portuguese counterparts, except for the census of commercial organizations; the Spanish results are robust to excluding this indicator.

Annex 2 provides further details on indicator construction, data sources, and additional variables requiring more extensive discussion, including mediators and post-treatment outcomes.

5. Empirical results

Table 1 reports OLS estimates from regressions of the three 2019 electoral outcomes on wartime exposure. Across specifications, the estimated relationships are negative and remain statistically significant after the inclusion of the full set of control variables.

Table 2 presents the corresponding IV estimates. Greater wartime exposure is associated with lower political polarization—on both measures—and lower support for populism. The estimates remain negative across specifications and across the three outcomes, indicating that the findings are not sensitive to a particular set of controls or to a single measure of political polarization. The IV estimates are larger than the corresponding OLS estimates, which may reflect selection bias resulting from some locations experiencing more battles because of ideological factors rather than logistical conditions, as well as potential confounding from later conflicts, such as the 1936–1939 Civil War. Both factors could be associated with higher polarization, all else equal, and may therefore help explain the smaller magnitude of the OLS estimates. This interpretation is further supported by the results of the Hausman endogeneity tests reported in Table A1 of Annex 1.

The first-stage estimates show a strong positive relationship between the instrument and the number of battles, with F-statistics above conventional weak-instrument thresholds in all specifications. Taken together, the results indicate a robust negative relationship between exposure to Carlist Wars battles and contemporary political polarization, as well as support for populism.

Table 1: Association between exposure to Carlist Wars battles and contemporary electoral outcomes.

	(1)	(2)	(3)	(4)
Dependent Variable				
Panel A: Political Polarization Index, RQ (2019)				
War Events (Count)	-0.008*** (0.001)	-0.006*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
Panel B: Dalton Polarization (2019), [Standardized]				
War Events (Count)	-0.039*** (0.005)	-0.008* (0.004)	-0.010** (0.004)	-0.007* (0.004)
Panel C: Populism Index (2019), [Standardized]				
War Events (Count)	-0.143*** (0.031)	-0.120*** (0.029)	-0.078*** (0.023)	-0.078*** (0.023)
Controls				
Instrument Components (levels and pairwise interactions)	No	Yes	Yes	Yes
Economic Indicators	No	No	Yes	Yes
Societal Indicators	No	No	No	Yes
Observations	5,312	5,312	5,100	5,100

Notes: Coefficients and standard errors are rounded to three decimal places. Each panel contains the results for a different dependent variable. Columns differ in the set of controls included. War Events are the number of times the grid cell was within a 25 km radius of a battle from the Carlist Wars. Instrument components are the levels and pairwise interactions of ruggedness, distance to the closest Crown naval arsenal and mule road density. Economic indicators are urbanization, irrigation and population in 1830, maximum caloric suitability, market potential in 1830 using time of transportation as cost, number of commercial organizations by 1830 divided by population, distance to the nearest port and density of royal, cart and post roads. Societal indicators are the average number of days elapsed between baptism date and birth date in each baptism place, the standard deviation of the days elapsed between baptism date and birth date in each baptism place, the average cosine similarity in each baptism location between the geographic distribution of the surnames of the father and the mother of each baptized child, and the distance from the grid centroid to the closest baptism location. All control variables are standardized. Standard errors, clustered at the NUTS-3 level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 2: Results from instrumental variable estimations.

	(1)	(2)	(3)	(4)
Dependent Variable (Second Stage)				
Panel A: Political Polarization Index, RQ (2019)				
War Events (Count)	-0.011** (0.005)	-0.016** (0.007)	-0.024*** (0.007)	-0.024*** (0.007)
Panel B: Dalton Polarization (2019), [Standardized]				
War Events (Count)	-0.436** (0.187)	-0.488** (0.221)	-0.201*** (0.078)	-0.188** (0.077)
Panel C: Populism Index (2019), [Standardized]				
War Events (Count)	-0.414* (0.241)	-0.072* (0.044)	-0.158** (0.066)	-0.159** (0.070)
First Stage				
Instrument [Standardized]	0.481*** (0.040)	0.496*** (0.057)	0.780*** (0.063)	0.721*** (0.063)
First-stage F-stat	142.700	53.580	43.990	41.480
Controls				
Instrument Components (levels and pairwise interactions)	No	Yes	Yes	Yes
Economic Indicators	No	No	Yes	Yes
Societal Indicators	No	No	No	Yes
Observations	5,312	5,312	5,100	5,100

Notes: Coefficients and standard errors are rounded to three decimal places. Coefficients in the “First Stage” report the effect of the instrument on War Events. The first-stage F-statistic is the Kleibergen–Paap F-statistic. In the second stage, each panel contains the results for a different dependent variable. In both stages, columns differ by the set of controls included. War Events are the number of times the grid cell was within a 25 km radius of a battle from the Carlist Wars. Instrument components are the levels and pairwise interactions of ruggedness, distance to the closest Crown naval arsenal and mule road density. Economic indicators are urbanization, irrigation and population in 1830, maximum caloric suitability, market potential in 1830 using time of transportation as cost, number of commercial organizations by 1830 divided by population, distance to the nearest port and density of royal, cart and post roads. Societal indicators are the average number of days elapsed between baptism date and birth date in each baptism place, the standard deviation of the days elapsed between baptism date and birth date in each baptism place, the average cosine similarity in each baptism location between the geographic distribution of the surnames of the father and the mother of each baptized child, and the distance from the grid centroid to the closest baptism location. All control variables are standardized. Standard errors, clustered at the NUTS-3 level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. Validity of the empirical strategy

I next provide additional evidence on the validity of the empirical strategy. I first examine whether the instrument predicts outcomes in settings where the proposed wartime mechanism should not operate. I then assess the sensitivity of the main results to potential violations of the exclusion restriction.

6.1. Predicting the geographical distribution of other wars

A key condition for instrument validity is that it not systematically predict other historical conflicts whose effects could be confounded with those of the Carlist Wars. I assess this empirically by testing whether the instrument predicts the geographic distribution of battles during the Peninsular War (1808–1814) or the Spanish Civil War (1936–1939).

The Peninsular War, triggered by the Napoleonic invasion and occupation of Spain, provides a useful placebo because it involved similar military technology but a different logistical mechanism. Guerrilla warfare was carried out mainly by locally organized irregulars—peasants, townspeople, former soldiers, and local militias—against French occupation forces, rather than as an insurgency the Spanish Crown sought to suppress (Tone, 1994; Esdaile, 2004; Scotti-Douglas, 2005). Formal Spanish military efforts relied primarily on regular armies and allied conventional campaigns, while French forces depended on reinforcements and supplies from France and the Napoleonic empire (Gates, 2001; Esdaile, 2003). Therefore, although guerrilla-suitable terrain may predict exposure to Peninsular War violence, there is no clear reason to expect such exposure to vary with distance to the Crown naval arsenals, since—unlike in the Carlist Wars—these guerrillas were not engaged in conflict with Crown forces.

The Spanish Civil War (1936–1939) is also informative as a placebo test because the relevant technological constraints differed substantially. By then, combat relied on modern military technologies, including tanks and aircraft, which likely reduced the importance of the logistical conditions that had shaped the distribution of conflict during the Carlist Wars.

The results in Table 3 show that the instrument does not predict the geographical distribution of battles in these other conflicts. This evidence supports the interpretation that the instrument captures logistical features specific to the Carlist Wars, in which the Crown sought to suppress an opponent that relied on guerrilla warfare. These features were absent from the other major conflicts that occurred after the creation of the Crown naval arsenals, where the instrument is consistently a poor predictor of battle locations.

Table 3: Predictive power of the instrumental variable in other conflicts in Spain

	(1)	(2)	(3)	(4)
Dependent Variable				
Panel A: Peninsular War (1808-1814) Events (Count)				
Instrument [Standardized]	-0.064 (0.044)	-0.233 (0.152)	-0.282 (0.169)	-0.255 (0.180)
Panel B: Civil War (1936-1939) Events (Count)				
Instrument [Standardized]	0.011 (0.021)	0.045 (0.029)	0.041 (0.032)	0.045 (0.032)
Controls				
Instrument Components (levels and pairwise interactions)	No	Yes	Yes	Yes
Economic Indicators	No	No	Yes	Yes
Societal Indicators	No	No	No	Yes
Observations	5,313	5,313	5,100	5,100

Notes: Coefficients and standard errors are rounded to three decimal places. Each panel contains the results for a different dependent variable. Events for each conflict are defined as the number of times a grid cell was within a 25 km radius of a battle during the Peninsular War or the Civil War. Instrument components are the levels and pairwise interactions of ruggedness, distance to the closest Crown naval arsenal and mule road density. Economic indicators are urbanization, irrigation and population in 1830, maximum caloric suitability, market potential in 1830 using time of transportation as cost, number of commercial organizations by 1830 divided by population, distance to the nearest port and density of royal, cart and post roads. Societal indicators are the average number of days elapsed between baptism date and birth date in each baptism place, the standard deviation of the days elapsed between baptism date and birth date in each baptism place, the average cosine similarity in each baptism location between the geographic distribution of the surnames of the father and the mother of each baptized child, and the distance from the grid centroid to the closest baptism location. All control variables are standardized. Standard errors, clustered at the NUTS-3 level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2. Liberal Wars in Portugal

The Portuguese Liberal Wars (1828–1834) provide a useful setting for assessing the validity of the instrument. Although the Portuguese and Spanish cases were similar in several important respects, warfare operated quite differently in the two contexts. If the instrument affects long-term political outcomes through channels other than the logistical mechanisms discussed in relation to the Carlist Wars, it should yield similar estimated effects in Portugal, even though the military conditions central to the Spanish case were absent or less salient there.

Like the Carlist Wars, the Portuguese conflict was triggered by a dynastic succession crisis, pitting the liberal constitutionalist claims associated with Pedro IV and Maria II against the absolutist claims of Miguel I (Paquette, 2013; Napier, 1836). More broadly, both Portugal and Spain were confronting the weakening and reconfiguration of their Atlantic empires during the nineteenth century (Monteiro, 2023;

Paquette, 2009, 2013). The two countries also shared a long and intertwined history within the Iberian Peninsula. For example, large parts of present-day Spain and Portugal were incorporated into al-Andalus after the Muslim conquest of 711 and were later connected to the Umayyad Emirate and Caliphate of Córdoba, as well as to subsequent Islamic polities in Iberia (Kennedy, 1996). The available data also make the comparison feasible: the road network is documented for the Iberian Peninsula as a whole, and Portugal had military structures equivalent to Spain’s Crown naval arsenals.

Despite these similarities, the Portuguese conflict was fought predominantly through conventional military operations and had a less pronounced guerrilla component (Lawrence, 2019; Napier, 1836). The warfare dynamics central to the Spanish case were therefore substantially less important in Portugal.

Table 4 shows that the corresponding pattern does not emerge following the Portuguese Liberal Wars. This finding suggests that the main results are unlikely to be driven by alternative channels linking the instrument to the outcome. The instrument is only weakly associated with political polarization and does not strongly predict battle exposure or polarization through that channel.

Table 4: Placebo test on the Liberal Wars (1828-1834) in Portugal.

	(1)	(2)	(3)	(4)	(5)	(6)
	Reduced Form		IV Estimation			
			Dependent Variable			
			Political Polarization Index, RQ			
War Events (Count)			0.037*	0.005	0.004	0.009
			(0.020)	(0.007)	(0.016)	(0.014)
			First Stage			
Instrument [Standardized]	-0.013**	-0.007	-0.399*	-1.639	-0.552	-0.716
	(0.005)	(0.008)	(0.210)	(1.009)	(0.568)	(0.606)
First-stage F-stat			3.596	0.742	1.667	11.760
	Controls					
Instrument Components (levels and pairwise interactions)	Yes	Yes	No	Yes	Yes	Yes
Economic Indicators	Yes	Yes	No	No	Yes	Yes
Societal Indicators	Yes	Yes	No	No	No	Yes
Country	Spain	Portugal	Portugal	Portugal	Portugal	Portugal
Year Dep. Var	2019	2021	2021	2021	2021	2021
Observations	5,100	966	1,007	1,007	966	966

Notes: Coefficients and standard errors are rounded to three decimal places. War Events are the number of times the grid cell was within a 25 km radius of a battle from the Portuguese Liberal Wars. The first-stage F-statistic is the Kleibergen–Paap F-statistic. Instrument components are the levels and pairwise interactions of ruggedness, distance to the closest Crown naval arsenal and mule road density. Economic indicators are urbanization, irrigation and population, maximum caloric suitability, market potential using time of transportation as cost, distance to the nearest port and density of royal, cart and post roads. Societal indicators are the average number of days elapsed between baptism date and birth date in each baptism place, the standard deviation of the days elapsed between baptism date and birth date in each baptism place, and the distance from the grid centroid to the closest baptism location. All control variables are standardized. Standard errors, clustered at the NUTS-3 level, are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

6.3. Conley-Hansen-Rossi bounds

The previous subsections suggest that the instrument does not predict the distribution of warfare in conflicts where the military asymmetry that makes the instrument relevant was absent. They also suggest that the instrument does not have a direct effect on outcomes in other similar contexts. Nonetheless, in this section I use Conley, Hansen, and Rossi (2012) to assess how large a violation of the exclusion restriction would have to be to explain away the results. Thus, consider equations [6] and [7]:

$$[6] \text{ } War_j = \alpha_1 + \gamma Z_j + \Omega_1 X_j^* + \varepsilon_{1j}$$

$$[7] Y_j = \alpha_2 + \beta War_j + \theta Z_j + \Omega_2 X_j^* + \varepsilon_{2j}$$

Here, X_j^* is the vector of controls, also including the lower-order terms of the variables used to construct the instrument, Z_j is the instrumental variable and the rest of the variables follow the same notation outlined in Section 4.2.1. Then, consider the reduced-form equation:

$$[8] Y_j = \alpha_3 + \delta Z_j + \Omega_3 X_j^* + \varepsilon_{3j}$$

The estimated coefficient will converge in probability to:

$$\hat{\delta} \xrightarrow{plim} \beta\gamma + \theta$$

Given this, one can iterate over different values of θ until the confidence interval for $\hat{\beta}$ includes zero. This procedure yields the maximum value of the direct effect (θ) that would be required for the violation of the exclusion restriction to explain away the results. Table 5 shows that the direct effect of the instrument would have to account for between 61% and 91% of the reduced-form effect for violations of the exclusion restriction to explain the findings.

Any violation of the exclusion restriction would have to explain most of the reduced-form effect, even after conditioning on extensive pre-treatment controls and the lower-order terms of the instrument components. This seems implausible given that the residual variation in the instrument has no clear theoretical link to the outcomes and is not evidently associated with other major developments in post-eighteenth-century Spain. The analysis suggests that such a direct effect would have to be more important than the prolonged civil war for which a documented mechanism exists.

Table 5: Sensitivity analysis with Conley-Hansen-Rossi bounds.

	(1)	(2)	(3)
	Dependent Variable		
	Political Polarization Index, RQ (2019)	Dalton Polarization (2019), [Standardized]	Populism Index (2019), [Standardized]
Instrument [Standardized]	-0.013** (0.005)	-0.115*** (0.038)	-0.330*** (0.070)
Direct Effect (for war effect non-significant)	-0.008	-0.079	-0.300
Direct Effect (% of Reduced Form)	61%	69%	91%
Controls	Yes	Yes	Yes
Observations	5,100	5,100	5,100

Notes: Coefficients and standard errors are rounded to three decimal places. All standardized control variables are included in the specifications. “Direct effect” corresponds to the value of θ under which the confidence interval of β includes 0. It is displayed in magnitude and as a % of the reduced form. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7. Robustness checks

The study involves several methodological decisions that may influence the conclusions. For instance, I use a 25 km radius around each battle location to proxy for areas exposed to the war, and I report results for 2019 as the main post-treatment year. These and other choices reflect the fact that there is no single, unambiguously correct way to operationalize exposure or to study these processes. I therefore assess whether the results are robust to a range of alternative methodological specifications.

Annex 1 reports the full set of estimates. Figure A1 shows that the results are stable across all municipal elections from 2011 onward, while Figure A2 shows that they are robust to alternative exposure thresholds ranging from 5 to 50 km. Figure A3 reports robustness checks allowing for flexible spatial autocorrelation, following Colella et al.’s (2023) refinement of Conley’s (1999) spatial HAC framework. The polarization results remain robust throughout. For populism, significance falls below the 5% level when spatial correlation is allowed beyond 50 km, but the estimates remain significant at the 10% level. Finally, Figure A4 shows that the results are robust when the outcome indicators are constructed using Spain’s general elections.

8. Mechanisms of persistence

8.1. Regional distinctiveness or political articulation?

As suggested by the evidence reviewed in Section 2, demands for greater local autonomy were neither clearly articulated before the Carlist Wars nor central to their outbreak (Henningsen, 1836; Zaratiegui, 1845). The conflict, however, politicized dimensions such as the *fueros* that later became relevant for peripheral and regionalist political movements (Carlos VII, 1872; Shih, 1998).

To study the mechanisms, I begin by exploring the extent to which pre-existing differences along dimensions later politicized by peripheral and regionalist movements explain the results, as opposed to the role of the Carlist Wars in politicizing them. These dimensions include cultural heritage, language, and administrative autonomy (Alonso, Gómez, & Cabeza, 2013). I therefore include fixed effects for historical polities that eventually lost administrative autonomy and for areas in which non-Castilian languages were traditionally spoken. This allows me to test whether the Carlist Wars politicized these dimensions more strongly in more affected areas within the same language area or historical polity.

Figures 5a and 5b map language areas and historical polities in 1492. The polity fixed effects capture grievances rooted in pre-Bourbon political institutions, such as the *fueros* of the Crown of Aragón, which were later politicized (Carlos VII, 1872) as authority became increasingly concentrated in the Crown of Castile during the eighteenth century. I use the 1492 boundaries because they closely approximate those preceding Bourbon centralization while allowing the former Emirate of Granada to retain its distinct institutional legacy²⁷.

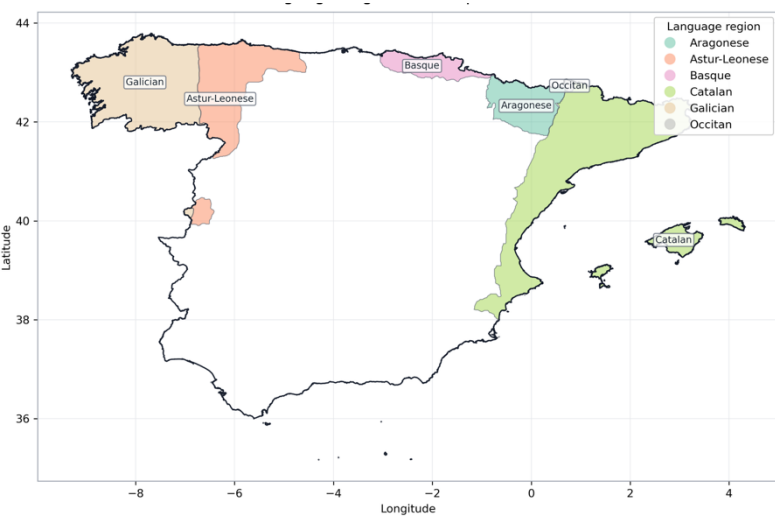
Table 6 presents the results from the main specification with these fixed effects. The results are robust to the inclusion of both language-area and historical-polity fixed effects. This suggests that, within areas with similar potential grievances against the centralized Crown, the electoral effects are stronger in locations more affected by the Carlist Wars. When I include a dummy for non-Castilian language instead of language-area fixed effects, the results are also robust. Together, these findings suggest that regional distinctiveness or the potential for grievances is not sufficient to produce the results. What seems to matter is not whether distinctiveness exists, but whether these dimensions become politicized and expressed within the political system—a process in which the Carlist Wars arguably played a role. This interpretation also helps explain why, for instance, Catalan regionalism plays an important role in

²⁷ The Emirate of Granada was incorporated into the Crown of Castile in 1492. Results are robust to treating this territory as part of the Crown of Castile for estimation purposes.

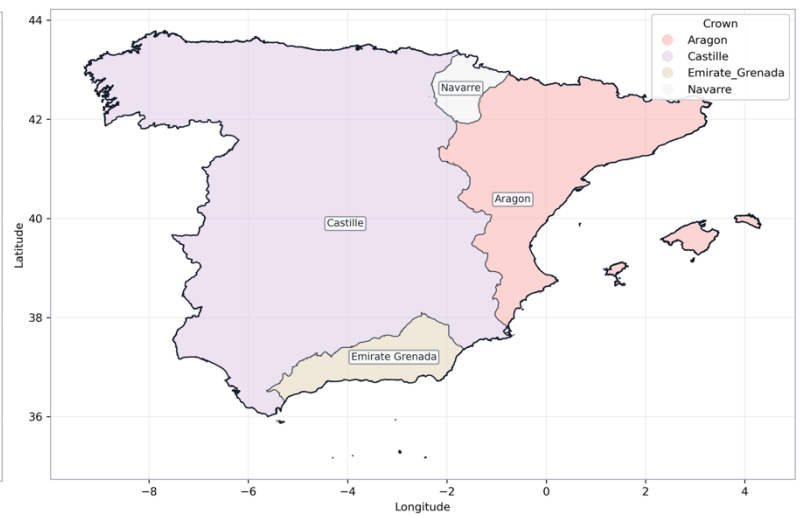
party competition in Spain but not in France, despite parts of France also lying within the historical area where the language developed (Balcells, 2013).

Figure 5: Linguistic map and polities by 1492.

a) Linguistic map.



b) Polities by 1492.



Notes: The language distribution of non-Castilian languages would have been stable for centuries before the 18th century (Enciclopedia Universal Ilustrada, 1923), although the degree of stability is subject to debate (Burgueño, 2002). I obtain the distribution from d-maps, which is consistent with García Mouton (2018), and the Linguistic Atlas of the Iberian Peninsula. For the Spanish polities by 1492, I use the Perry-Castañeda Library Map Collection hosted at UT Austin.

Table 6: IV results with polity and language zone fixed effects or non-Castilian area dummy.

	(1)	(2)	(3)	(4)	(5)
Dependent Variable (Second Stage)					
Panel A: Political Polarization Index, RQ (2019)					
War Events (Count)	-0.025*** (0.007)	-0.024*** (0.005)	-0.017** (0.007)	-0.018** (0.008)	-0.018** (0.007)
Panel B: Dalton Polarization (2019), [Standardized]					
War Events (Count)	-0.311** (0.132)	-0.178** (0.076)	-0.289** (0.139)	-0.408** (0.186)	-0.304** (0.139)
Panel C: Populism Index (2019), [Standardized]					
War Events (Count)	-0.170** (0.066)	-0.153** (0.069)	-0.189** (0.077)	-0.210*** (0.075)	-0.201*** (0.075)
Controls	Yes	Yes	Yes	Yes	Yes
Language Fixed Effects	Yes	No	No	Yes	No
(Non-Castilian Language=1)	No	Yes	No	No	Yes
Polity Fixed Effects	No	No	Yes	Yes	Yes
Observations	5,100	5,100	5,100	5,100	5,100

Notes: Coefficients and standard errors are rounded to three decimal places. Each column changes the fixed effects that are included. Controls include all variables from the preferred specification. All control variables are standardized. Standard errors, clustered at the NUTS-3 level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8.2. Instrumental variable estimation of post-treatment outcomes

This section examines how war exposure shaped historical outcomes in the decades after the conflict. Using instrument-predicted variation conditional on controls, I estimate the preferred specifications in Equations [4] and [5], replacing contemporary polarization and populism with post-war outcomes measured between 1876 and 1936.

Yearly data for the nineteenth century are unavailable, so I cannot estimate the effects of war exposure immediately following major battles or events. Moreover, many societal outcomes may take time to materialize. I therefore rely on indicators measured in the decades following the wars.

I first examine societal channels plausibly shaped by the war, as discussed in Section 3: social cohesion, political participation, collective action, political cleavages, and party formation (Blattman, 2009; Bellows & Miguel, 2009; Voors et al., 2012; Gilligan, Pasquale, & Samii, 2014; Lipset & Rokkan, 1967). Because data for this period are limited, the variables used here provide the broadest feasible set of proxies for these dimensions. The discussion below describes each indicator.

I organize the outcomes into three groups. First, I examine historical political outcomes from the late nineteenth century to 1936. Second, I use post-war analogues of the societal controls in the main specification to assess whether war exposure also shaped religiosity, family formation patterns, and local organization. Third, I examine post-1975 municipal elections to test whether related political channels persisted into the democratic period.

I use constituency-level election data from 1876 to 1936 to construct two indicators²⁸. First, using the party affiliation of elected members of Congress after each election, I compute the share of representatives from regional parties, defined as parties operating in specific regions rather than nationwide. Because such parties had emerged in all regions outside Madrid by 1936²⁹, this measure captures both their presence and their ability to convert local political projects into parliamentary representation. It therefore provides evidence on local political mobilization and the institutional translation of collective action into political power.

Second, using party vote shares from 1931 to 1936, I construct a measure increasing in the cross-cuttingness of cleavage structures along the left-right and center-periphery dimensions, following Taylor and Rae (1969). Comparable vote-share data are unavailable before 1931, when only elected representatives are observed.

Additionally, I compute post-war indicators that proxy societal channels using variables originally included as pre-war controls in the main specification. These include the Dalton indicator from time elapsed between birth and baptism; the spouse geographic similarity indicator; and the number of commercial organizations per capita in 1880 and 1916. The latter proxies non-electoral organizational and economic capacity, helping distinguish political mobilization from broader local organizational change. Because of data limitations, I cannot compute yearly indicators for these dimensions. In the case of baptisms, I pool all records collected between 1876 and 1920 and compute the indicators at the baptism-place level.

I also use data from all post-1975 municipal elections to construct two further indicators that aim to capture the same electoral dimensions as those measured up to 1936, allowing me to assess continuity

²⁸ Annex 2 provides details on all data sources used in this section, the coding procedures, and the construction of each indicator. Ideological position is based on Artola (1975) and on coalition-building patterns over time. These measures should also be interpreted in light of changes in the electorate: universal male suffrage was introduced in 1890, and women's suffrage in 1931.

²⁹ Annex 2 describes the procedure used to compile the broadest possible set of political parties founded before 1936 for which data are available.

in the channels under study. First, I compute the percentage of votes received by regional lists³⁰—which are not necessarily regionalist—to approximate the dimension captured by the percentage of congresspeople elected from regional parties before 1936. Second, I construct a measure that increases with the cross-cuttingness of cleavage structures, following Baldassarri and Gelman’s (2008) approach. I complement this measure by weighting it with issue salience, following the framework of Ansolabehere and Puy (2018), to address limitations of the baseline measure discussed in Annex 2. I do not use Taylor and Rae’s (1969)³¹ measure for this period because I operationalize cleavage positions and salience using data from the RMP, which provides continuous measures of parties’ positions on the left-right and center-periphery dimensions.

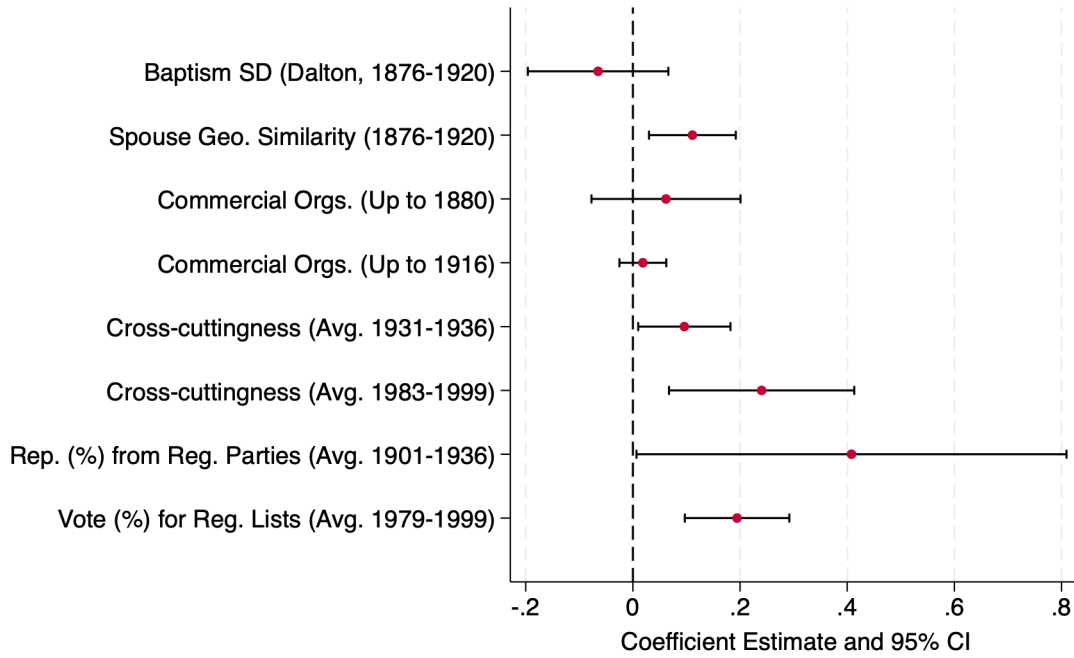
Figure 6 shows positive and statistically significant effects of war exposure on spouse geographic similarity, the share of congresspeople elected from regional parties from 1901³² onward, the vote share of regional lists after 1975, and cross-cuttingness both before 1936 and after 1975. By contrast, the effects on the Dalton indicator and on commercial organizations per capita in 1880 and 1916 are not statistically significant. Taken together, these findings support the interpretation that the war strengthened geographically rooted identities, increased affected regions’ capacity to articulate local political projects and secure representation through regional parties, and contributed to more multidimensional forms of political competition. The absence of effects on commercial organizations suggests that these patterns operated primarily through political channels rather than broader economic or organizational change.

³⁰ Annex 2 discusses how this is defined in post-1975 elections.

³¹ Taylor and Rae (1969) use categorical ideological indicators.

³² Regional parties first gained seats in Congress in 1901.

Figure 6: Effects on standardized post-war outcomes.



Notes: “Coefficient Estimate and 95% CI” reports the instrumental-variable estimate of the effect of war exposure on each outcome, along with its 95% confidence interval. “Baptism SD (Dalton, 1876-1920)” is the standard deviation of the difference between baptism date and birth date in baptism records in each baptism location from 1876 to 1920. “Spouse Geo. Similarity (1876-1920)” is the average between 1876 and 1920 in each baptism location of the cosine similarity between the geographical distribution of paternal and maternal surnames of baptized children based on the pre-war geographic distribution. “Commercial Orgs.” represents the number of commercial organizations up to the stated year (1880 and 1916) divided by population. “Cross-cuttingness (Avg. 1931-1936)” represents the XC indicator from Taylor and Rae (1969) based on vote shares and party ideology. “Cross-cuttingness (Avg. 1983-1999)” represents the cross-cutting indicator for continuous dimensions based on Baldassarri & Gelman (2008). The ideological dimensions are constructed from the averaged RILE and center-periphery positions from the Regional Manifestos Project up to the year 2000. Because of data limitations, this indicator could only be constructed from 1983 onwards. “Rep. (%) from Reg. Parties (Avg. 1901-1936)” is the average percentage of congresspeople by constituency who are from a regional party. The indicator used is the average of this measure in each constituency during the period. “Vote (%) for Reg. Lists (Avg. 1979-1999)” represents the percentage of votes for electoral lists that appear in at most four autonomous communities. The indicator used is the average of this measure in each municipality during the period. Standard errors are clustered at the NUTS-3 level.

8.3. Mediation analyses

In the previous section, I found evidence that the treatment influenced several relevant outcomes in the decades following the war. To link these changes to contemporary outcomes, I use mediation analyses to assess whether they help identify channels through which the Carlist Wars may have shaped contemporary political polarization and populism. Specifically, I follow the “Causal Mediation Analysis” approach proposed by Dippel et al. (2020), which leverages variation introduced by the instrument to isolate the role of the war from other factors that may have shaped the mediators. This strengthens the attribution of the proposed mechanisms to the war itself.

Because I have one instrument and several potential mediators, I apply the IV-mediation framework as a sequence of single-mediator exercises. I therefore interpret the results as tests of the plausibility of the proposed mechanisms, rather than as a precise numerical decomposition of the overall effect, since the mediators may be conceptually related or operate in tandem. Considering multiple potential channels, including important competing explanations, allows me to distinguish mechanisms supported by the data from less supported alternatives.

I complement this analysis with Kwon and Roth's (2026) test of the sharp null of full mediation, applying it to the mechanisms that receive the strongest empirical support in the preceding analysis. Rejection of the null for supported channels would suggest that other mechanisms not considered in the analysis may explain the relationship between war exposure and contemporary political outcomes.

Kwon and Roth's (2026) test is developed for categorical mediators, whereas the mediators considered here are continuous. Applying the test therefore requires discretization, which the authors caution against unless the mediator can reasonably be assumed not to affect the outcome within each category, an assumption that is unlikely to hold in this setting. To mitigate this concern, I discretize the mediators into tertiles and quartiles and test whether the conclusions are robust. Nonetheless, given these limitations, I interpret the results as secondary, diagnostic evidence of whether channels not closely correlated with the more strongly supported mechanisms may still account for an important share of the treatment effect.

Table 7 supports two mechanisms linking the war to contemporary political polarization and populism: the emergence of local or regional political projects and the institutionalization of cross-cutting cleavages within the party system. There is some evidence of an increased role in preserving geographically rooted lineages, but it is weaker. Indicators proxying for religious uniformity and economic organizational capacity receive little support. Table 8 reports p-values from the Kwon and Roth (2026) test. Full mediation is not rejected for the supported mechanisms, although evidence is again weaker for the lineage indicator. This suggests limited evidence that an additional, weakly related channel not considered in the analysis helps explain the results.

Overall, the findings from the analysis of mechanisms are consistent with the theoretical framework presented in Section 3. In affected regions, competition along the center-periphery axis appears to cut across national political divisions, as also reflected in the greater relevance of regional political projects

regardless of their ideological orientation. This appears to explain why political polarization and support for populist parties are lower.

Table 7: IV mediation analyses.

	(1)	(2)	(3)
	Dependent Variable		
	Political Polarization Index, RQ (2019)	Dalton Polarization (2019), [Standardized]	Populism Index (2019), [Standardized]
	Total Effect		
War Events (Count)	-0.024*** (0.007)	-0.188** (0.077)	-0.159** (0.070)
Effect Through Mediators			
Baptism SD (Dalton, 1876-1920), [Standardized]	-0.060 (0.204)	-0.531 (1.845)	-0.237 (0.831)
Spouse Geo. Similarity (1876-1920), [Standardized]	-0.023* (0.012)	-0.199* (0.115)	-0.089** (0.041)
Commercial Organizations per Capita (up to 1916), [Standardized]	-0.020 (0.033)	-0.180 (0.327)	-0.081 (0.168)
Vote Share (%) for Regional Lists (Avg. 1979-1999), [Standardized]	-0.034*** (0.012)	-0.302*** (0.095)	-0.135*** (0.039)
Cross-cuttingness (Avg. 1983-1999), [Standardized]	-0.020*** (0.007)	-0.175*** (0.052)	-0.078** (0.038)

Notes: Coefficients and standard errors are rounded to three decimal places. “Baptism SD (Dalton, 1876-1920)” is the standard deviation of the difference between baptism date and birth date in baptism records in each baptism location from 1876 to 1920. “Spouse Geo. Similarity (1876-1920)” is the average between 1876 and 1920 in each baptism location of the cosine similarity between the geographical distribution of paternal and maternal surnames of baptized children based on the pre-war geographic distribution. “Commercial Organizations” represents the number of commercial organizations up to 1916 divided by population. “Vote share (%) for regional lists (Avg. 1979-1999)” represents the percentage of votes for electoral lists that appear in at most four autonomous communities. The indicator used is the average of this measure in each municipality during the period. “Cross-cuttingness (Avg. 1983-1999)” represents the cross-cutting indicator for continuous dimensions based on Baldassarri & Gelman (2008). The ideological dimensions are constructed from the averaged RILE and center-periphery positions from the Regional Manifestos Project up to the year 2000. Because of data limitations, this indicator could only be constructed from 1983 onwards. All control variables are standardized. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: P-values from the Kwon and Roth (2026) sharp mediation test.

Mediators	Kwon-Roth p-values (CS)	
	Tertiles	Quartiles
Spouse Geo. Similarity (1876-1920)	0.056	0.171
Vote share (%) for Regional Lists (Avg. 1979-1999)	0.549	0.606
Cross-cuttingness (Avg. 1983-1999)	0.404	0.422
All Mediators	0.615	0.796

Notes: P-values are rounded to three decimal places. “Tertiles” and “Quartiles” refer to the discretization methods applied to the mediators. “CS” denotes the Cox-Shi method, as recommended by Kwon and Roth (2026). All control variables and mediators are standardized.

9. Alternative channels and mechanism attribution

Early industrialization, educational expansion, and migration provide alternative explanations for the political patterns documented above. Industrialization may explain why regional interests emerged in some regions but not others, independently of the Carlist Wars. Educational expansion, in turn, may explain how these interests were consolidated and transmitted across generations. Migration raises a distinct measurement-related concern, which I address below, and could confound the results.

Because these processes unfolded after the wars, directly controlling for variables that measure them would risk introducing bad-control bias. I therefore examine whether they mediate the IV estimates. For industrialization and migration, this analysis also speaks to the exclusion restriction: while both could plausibly be channels through which the Carlist Wars affected later political outcomes, significant mediation could also suggest that the instrument captures broader historical processes that evolved independently of the wars³³. For educational expansion, the purpose is different. The mediation analysis helps assess whether the persistence of electoral effects is better explained by the patterns of party competition examined in the previous section, or by schooling as an alternative channel through which peripheral identities and interests were transmitted over the long run.

³³ The mediation analysis in Dippel et al. (2020) estimates the effect of the mediator on the outcome, conditional on the treatment, by instrumenting the mediator while controlling for the treatment.

9.1. Early industrialization

A body of literature in Spanish history links the emergence of regional party structures and stronger competition along the center-periphery axis to the early industrialization of certain regions. Early industrialization may have facilitated stronger bourgeois organization, denser associational networks, and less fragmented political vehicles (Gellner, 1983; Fusi, 1994).

I measure industrialization in two ways. The first measure follows Parejo (2006) and uses the share of the population employed in industrial or manufacturing sectors in the 1860 census³⁴. The second follows Beltrán Pérez (1999) and uses provincial estimates of per capita tax revenue from industrial activities. To obtain grid-cell variation, I project the provincial estimates onto cells using geographic and economic predictors, constraining the projected values to reproduce the provincial averages reported for 1913 and 1929³⁵.

9.2. Timing of mass literacy expansion

An alternative channel of persistence concerns the timing of mass literacy. Building on Darden’s (2013) theory of the “scholastic revolution,” Balcells (2013) argues that the greater contemporary salience of Catalan interests in Spanish Catalonia than in French Catalonia can be partly explained by differences in the timing of mass-literacy expansion in Spain and France. Spain’s limited capacity to establish a nationwide nationalizing school system delayed the scholastic revolution until the early twentieth century (Balcells, 2013; Linz, 1973). As a result, mass literacy expanded at a time when regional political mobilization was already salient (Balcells, 2013; Hroch, 2000), facilitating the consolidation and intergenerational transmission of regional identities through schooling.

This logic can be extended beyond Catalonia (Darden & Grzymala-Busse, 2006)³⁶. Educational expansion may have reinforced perceived differences with the center and helped consolidate center-periphery cleavage structures in other regions. This argument is consistent with research emphasizing how mass schooling transmits durable political narratives and national identities across generations

³⁴ Parejo (2006) uses the share of workers in manufacturing, mining, and transport-related occupations. I measure this using the share of manufacturers, industrial workers, miners, railway employees, and laborers, which I group into an “industrial core.” For robustness, I also construct an alternative metric that includes forms of proto-industrial work, such as artisans and architects. The results are robust to this alternative measure.

³⁵ The determinants include ruggedness, urbanization, population, distance to port, railroad density and stations measured in the closest available year, and Parejo’s (2006) industrialization indicator.

³⁶ In fact, the cross-national evidence for this mechanism comes from Darden and Grzymala-Busse (2006), whereas Balcells (2013) demonstrates its applicability to the Catalan case.

(Darden & Grzymala-Busse, 2006; Darden, 2013), while complementing broader work on schooling as an instrument of state-building and political socialization (Paglayan, 2022).

I test this channel using changes in the share of the population able to read and write. Based on the corresponding censuses, I calculate literacy growth for 1860–1900, 1900–1920, and 1920–1940.

9.3. Contemporary migration as a confounder

Migration raises a different concern related to the construction of the outcome. The populism index partly captures exclusionary appeals that distinguish “the people” from immigrants, and anti-immigrant attitudes are an important source of populist support (Norris & Inglehart, 2019; Dal Bó et al., 2023; Guriev & Papaioannou, 2022). The estimates could therefore reflect lower contemporary exposure to immigrants in areas affected by the wars rather than the historical cleavage-formation mechanism proposed above.

I assess this using [ineAtlas](#)’s foreign-born population share. Since the measure is contemporaneous with the political outcomes, it tests whether current migration exposure explains the estimated effect.

Table 9 shows no significant mediation through industrialization, literacy expansion, or contemporary migration. This does not imply that these processes were historically unimportant. Rather, insofar as the effects of the Carlist Wars do not appear to operate through channels that could independently generate similar outcomes, this finding reduces concerns about violations of the exclusion restriction. It also suggests that the evidence is stronger for the party competition mechanisms documented in the previous section than for an effect of the Carlist Wars operating through industrialization, educational expansion, or migration.

Table 9: Mediation analysis with potential confounders.

	(1)	(2)	(3)
	Dependent Variable		
	Political Polarization Index, RQ (2019)	Dalton Polarization (2019), [Standardized]	Populism Index (2019), [Standardized]
	Total Effects		
War Events (Count)	-0.024*** (0.008)	-0.188** (0.077)	-0.159** (0.070)
Effects Through Mediators			
Industrialization (Occupation) 1860, [Standardized]	-0.031 (0.256)	-0.270 (2.282)	-0.121 (0.997)
Industrialization (Tax) 1913, [Standardized]	-0.032 (0.022)	-0.286 (0.250)	-0.128 (0.136)
Industrialization (Tax), 1929, [Standardized]	-0.045 (0.040)	-0.399 (0.434)	-0.178 (0.241)
% Foreign Born [Standardized]	-0.013 (0.009)	-0.184 (0.126)	-0.082 (0.085)
Literacy Growth Rate (1860-1900), [Standardized]	-0.013 (0.010)	-0.149 (0.093)	-0.383* (0.229)
Literacy Growth Rate (1900-1920), [Standardized]	-0.006 (0.044)	-0.064 (0.508)	-0.164 (1.309)
Literacy Growth Rate (1920-1940), [Standardized]	-0.014 (0.010)	-0.162 (0.122)	-0.419 (0.351)

Notes: Coefficients and standard errors are rounded to three decimal places. “Industrialization (Occupation)” corresponds to the percentage of labor employed in the industrial sector, following Parejo (2006). “Industrialization (Tax)” is the metric used to measure industrialization by Beltrán Pérez (1999). “% Foreign Born” is the percentage of people who reported being born outside Spain in the 2021 census. Literacy growth rate is the rate of increase in the percentage of the population that reported being able to read and write in successive censuses. All control variables are standardized. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

10. Limitations, external validity, and future research

The study has several limitations that should be considered when interpreting the findings and the mechanisms that explain them. These limitations also point to avenues for future research.

First, the study is limited by its quantitative focus. While this approach allows for systematic analysis and strengthens attribution of later outcomes to the conflict, it cannot fully reconstruct the local micro-processes through which the documented effects occurred. For example, I cannot observe how communities interpreted the Crown’s military presence, how grievances became politically organized, or how local actors translated wartime experiences into party formation, electoral mobilization, or other

forms of collective action. Similarly, while political parties compete differently at the local level through targeted pamphlets, rallies, and other strategies that respond to local cleavage structures, I cannot observe how these forms of competition manifested over time. Instead, I infer these mechanisms from quantitative patterns interpreted through the theoretical framework and historical accounts. A deeper exploration of how these processes unfolded would require qualitative methods, such as primary-source analysis and historical process tracing across contexts in which political parties and organizations developed. The large body of available primary sources makes such work possible and constitutes an important avenue for future research.

Second, the study focuses on cross-cuttingness across only two axes of political conflict: left-right and center-periphery, given their importance in the Spanish case (Alonso, Gómez, & Cabeza, 2013). However, political polarization and cross-cuttingness could also be explored along other ideological dimensions beyond the cleavages identified by Lipset and Rokkan (1967). Whether similar patterns emerge with other dimensions of political conflict remains a topic for future research, including debates related to gender, LGBTIQ+ rights, and other cultural topics.

Finally, the study explains variation in local political outcomes by leveraging the substantial cross-sectional heterogeneity in patterns of political competition within Spain. It does not examine these conflicts at the level of Spain as a national unit, nor does it compare Spain with other countries. As a result, whether the findings can be extrapolated to larger polities remains an open question.

In this last respect, the theoretical framework discussed in Section 3 does not require these processes to operate locally, suggesting broader applicability. From this perspective, when historical processes freeze cross-cutting cleavages into party systems, their effects on polarization and populism should depend on whether those cleavages are represented nationally, regionally, or locally. If cross-cutting cleavages are incorporated into national party systems, they may be associated with lower-than-expected levels of polarization and populism at the national level. If they are represented primarily by regional parties, they may instead be associated with lower-than-expected levels of polarization and populism locally or regionally, without necessarily producing the same pattern nationally. The latter scenario resembles the Spanish case examined here. Nonetheless, these claims remain informed hypotheses that require further research.

11. Conclusion

This paper has argued that one potential channel explaining why some societies become more polarized or susceptible to populism is how historical conflicts during the expansion of mass suffrage shaped long-term patterns of political competition and, in turn, parties' incentives to intensify or moderate polarization. Drawing on foundational political science theories and using the Carlist Wars in nineteenth-century Spain as a setting, I find that greater exposure to the conflict is associated with lower political polarization and weaker support for parties using populist rhetoric in contemporary municipal and general elections. These findings are robust across electoral cycles, alternative specifications, placebo exercises, sensitivity analyses, and tests of the main competing explanations.

Evidence on mechanisms suggests that the effects operate through greater salience of a cross-cutting center-region cleavage and stronger support for regional and local political projects. Insofar as this additional axis of political competition does not correlate perfectly with the left-right divide, parties have weaker incentives to intensify polarization by adopting extreme positions on one axis, since doing so may cost them voters on the other. I find evidence that this also translates into weaker support for parties employing populist rhetoric, consistent with literature linking political polarization to populism.

These findings highlight an additional channel through which current global trends in political polarization and populism could be explored.

12. References

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1. Annex 1: Additional estimations and robustness checks

1.1. Hausman endogeneity tests

Table A1 reports p-values from Hausman tests of whether the OLS and IV estimates are asymptotically equivalent. Rejection of the null indicates systematic differences between the estimators and, conditional on instrument validity, supports the use of IV.

Table A1: P-values from Hausman endogeneity tests.

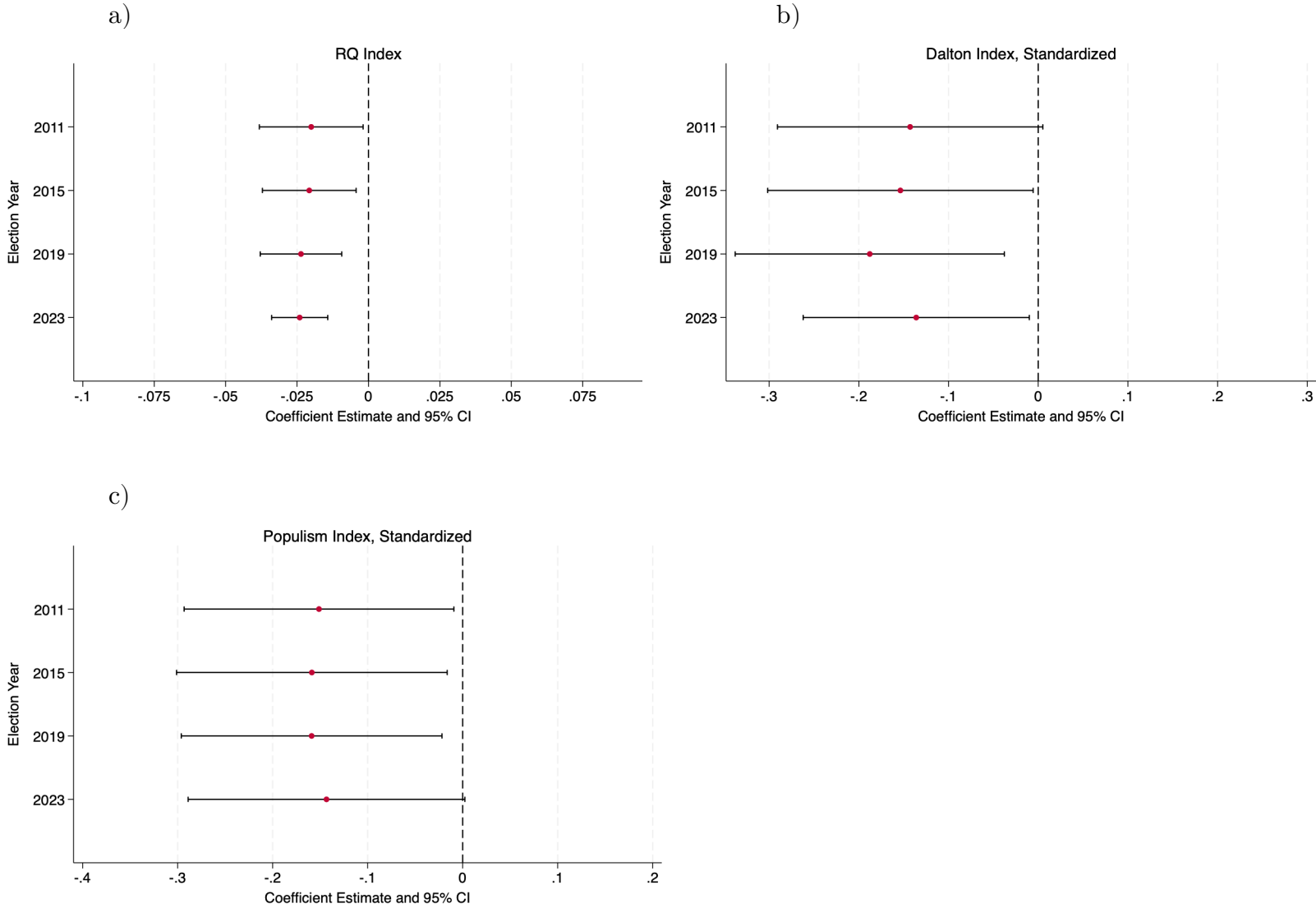
	(1)	(2)	(3)	(4)
Hausman endogeneity tests				
DEPENDENT VARIABLES	p-value			
Political Polarization Index, RQ (2019)	0.073	0.044	0.006	0.008
Dalton Political Polarization (2019) [Standardized]	0.001	0.000	0.007	0.013
Populism Index (2019) [Standardized]	0.014	0.067	0.025	0.027
Controls				
Instrument Components (levels and pairwise interactions)	No	Yes	Yes	Yes
Economic Indicators	No	No	Yes	Yes
Societal Indicators	No	No	No	Yes

Note: P-values are rounded to three decimal places.

1.2. Results on different municipal electoral cycles

Panels a), b), and c) of Figure A1 show the results of the main specification for electoral outcomes in municipal elections from 2011 onward for each of the dependent variables.

Figure A1: Instrumental variable estimation results on all municipal elections after 2010.

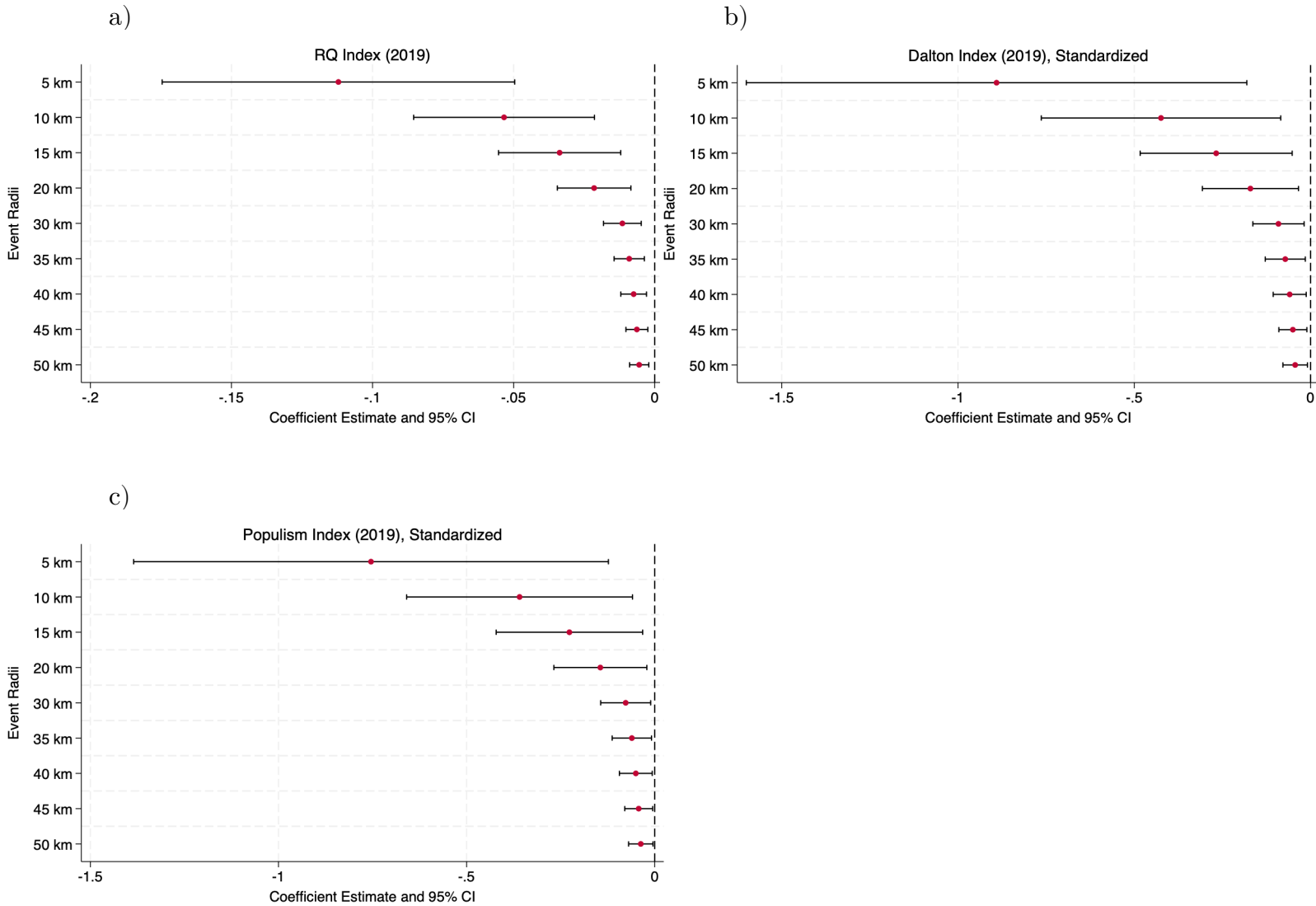


Notes: “Coefficient Estimate and 95% CI” reports the instrumental-variable estimate of the effect of war exposure on each outcome, along with its 95% confidence interval. The estimations include all control variables, and standard errors are clustered at the NUTS-3 level.

1.3. Different treatment definition

Figure A2 reports estimates in which exposure is defined as the number of times a grid cell falls within radii ranging from 5 to 50 km around each battle, for each dependent variable³⁷.

Figure A2: Instrumental-variable results under alternative treatment-definition radii.



Notes: “Coefficient Estimate and 95% CI” reports the instrumental-variable estimate of the effect of war exposure on each outcome, along with its 95% confidence interval. The estimations include all control variables, and standard errors are clustered at the NUTS-3 level.

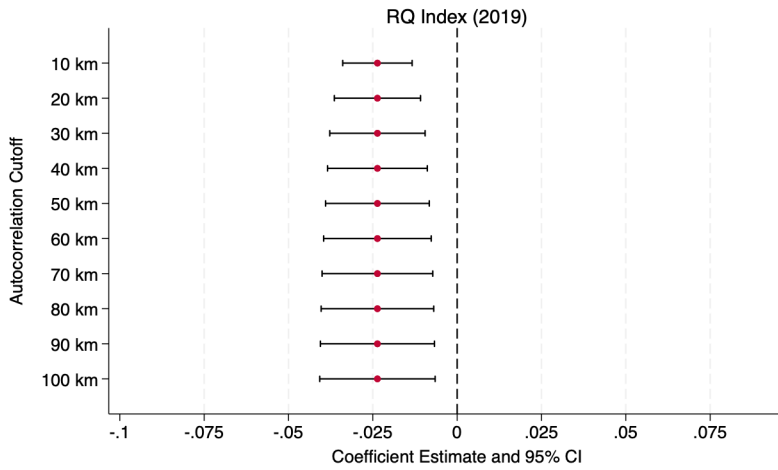
³⁷ This excludes 25 km, as this is the radius used in the main text.

1.4. Alternative forms of spatial autocorrelation

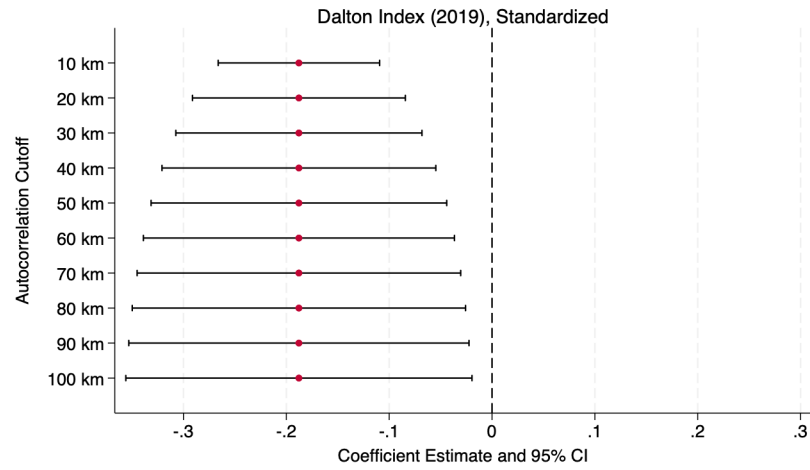
In the main specification, I report standard errors clustered at the NUTS-3 level. To assess sensitivity to spatial correlation across nearby grid cells, I also compute spatially corrected standard errors for alternative distance cutoffs. Following Colella et al.'s (2023) arbitrary-clustering approach, building on Conley's (1999) spatial HAC framework, I allow residual correlation among grid cells within alternative radii, up to 100 km. For 10×10 km cells, a 100 km radius already covers a broad local neighborhood—approximately 300 cells around each interior cell. Larger cutoffs would therefore imply dependence over a wide regional scale, equivalent to roughly 1/16 of Spain's area, rather than local spatial autocorrelation. Panels a)–c) of Figure A3 report these results.

Figure A3: Instrumental-variable results under alternative spatial-autocorrelation assumptions.

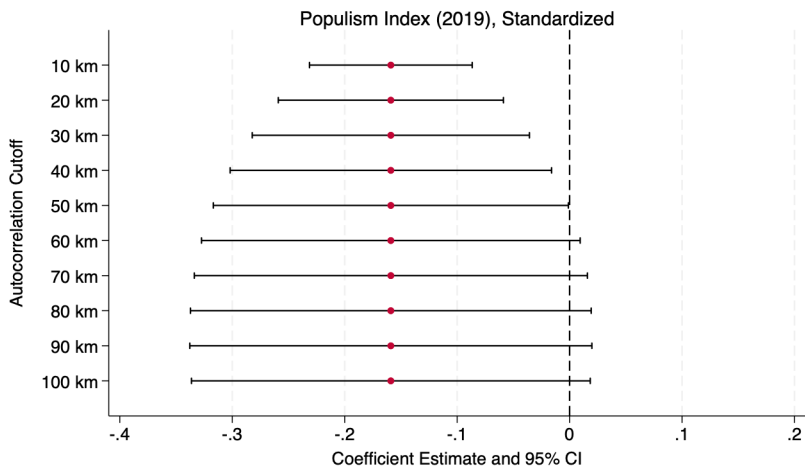
a)



b)



c)



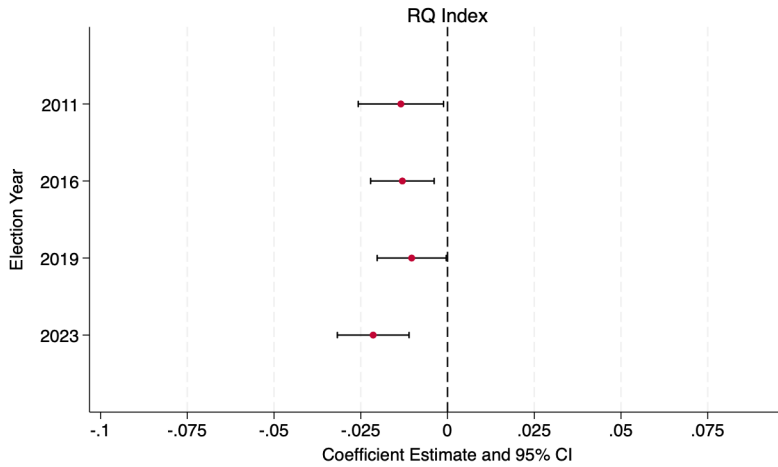
Notes: "Coefficient Estimate and 95% CI" reports the instrumental-variable estimate of the effect of war exposure on each outcome, along with its 95% confidence interval. The estimations include all control variables.

1.5. Results on general elections

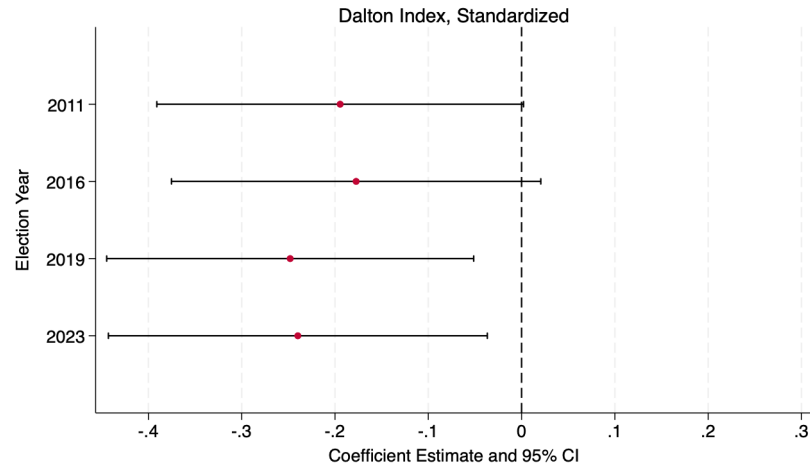
I assess whether the documented results also hold for general elections, while accounting for their limitations—particularly the 2015 election, which was repeated in 2016. The results, shown in Figure A4, are robust to election type, suggesting that party preferences are relatively stable across elections.

Figure A4: Instrumental variable estimation results on all general elections from 2011 onward.

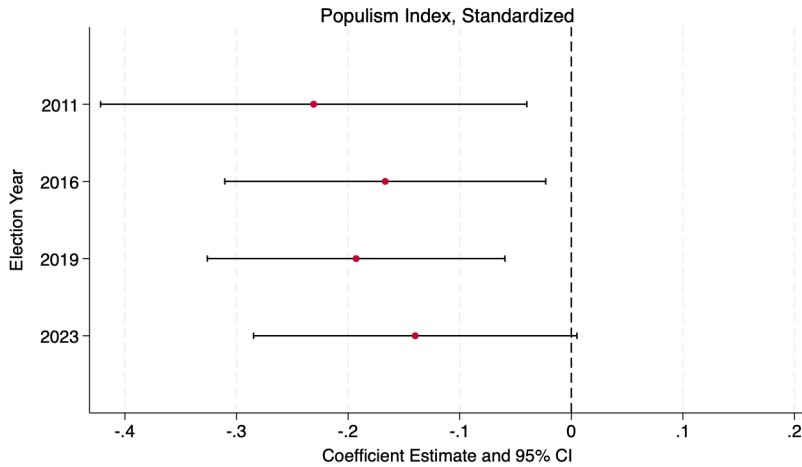
a)



b)



c)



Notes: “Coefficient Estimate and 95% CI” reports the instrumental-variable estimate of the effect of war exposure on each outcome, along with its 95% confidence interval. The estimations include all control variables, and standard errors are clustered at the NUTS-3 level.

2. Annex 2: Data annex

2.1. Construction of dependent variables

The formulas for each dependent variable are presented in the main text. This section provides additional details on data sources and other information not included there.

2.1.1. Data on party ideology

The party ideology data from the Regional Manifestos Project divide manifestos into quasi-sentences and assign each quasi-sentence to a single label based on its content. A sentence may contain multiple quasi-sentences; a quasi-sentence is the smallest communicative unit expressing content associated with one of the labels. This approach makes it possible to extract distinct communicative content from sentences that refer to more than one ideological dimension.

For each label, the dataset reports the percentage of all quasi-sentences corresponding to that topic. It also includes additional indicators based on these classifications, such as the RILE position, which places each party on the left-right spectrum, and the CP position, which places each party on the center-periphery dimension. The latter captures parties' positions on how much autonomy, authority, and responsibility in areas such as culture and education should be granted to regions vis-à-vis the central government. RILE saliency and CP saliency measure the importance of each dimension in the manifestos. For more details on the coding procedure and the complete set of variables and labels in the dataset, please check the [Regional Manifestos Project website](#).

As briefly discussed in the main text, the dataset classifies parties according to their regional variants. For example, in the case of the PP, the dataset includes separate indicators for regional branches such as PP Andalucía and PP Aragón. It also includes explicitly regional parties, such as Partido Andalucista, EH Bildu, and Chunta Aragonesista.

The objective is to classify parties ideologically, rather than to capture the specific content of a particular electoral cycle or election. Accordingly, the indicators are applied to both municipal and general elections and averaged over the period from 2010 onward.

2.1.2. Populism Indicator

The Regional Manifestos Project (RMP) does not contain a direct measure of populism. I therefore use principal component analysis (PCA) to identify a latent component underlying a set of manifesto indicators theoretically associated with populist discourse.

As discussed in the main text, the analysis does not seek to operationalize only a minimal definition of populism. Instead, it examines whether exclusionary, anti-institutional, and sovereignist forms of rhetoric co-vary in a manner consistent with the broader manifestations of populism identified in the literature (Mudde, 2004; Müller, 2016; Basile & Mazzoleni, 2020; Urbinati, 2019; Verzichelli, 2020; Guriev & Papaioannou, 2022). None of the indicators included below is treated as inherently or exclusively populist. The interpretation instead depends on whether they vary together as part of a common latent component. This operationalization is consistent with broader definitions of populism or how it tends to manifest itself (Guriev & Papaioannou, 2022).

The indicator is constructed from 16 RMP variables organized into three theoretical dimensions: (1) people-centric antagonism, (2) anti-institutionalism, and (3) sovereignism (protectionism and anti-globalization). These dimensions structure the selection and interpretation of the variables. They are constructed additively, following the dataset’s approach to indicators such as the RILE and CP positions. The dimensions are operationalized below, with dataset variable names shown in parentheses, e.g., “(c452).”

- (1) **People-centric antagonism:** Populist discourse presents “the people” as a morally unified community whose interests have been neglected or betrayed by a corrupt or self-serving “elite” (Mudde, 2004; Mudde & Rovira Kaltwasser, 2017). It often defines “the people” in exclusionary terms, distinguishing the supposedly authentic people from groups portrayed as outsiders or threats to social cohesion. This helps explain why some definitions of populism, for instance, emphasize xenophobic or anti-multiculturalist components (Guriev & Papaioannou, 2022).

Operationalization: People-centric antagonism is increasing in anti-multiculturalism (c608), welfare limitations for immigrants (c5051), limitations on education for immigrants (c5071), immigrants being a problem for law and order (c6051), and decreasing in pro-multiculturalism (c607), equal treatment of immigrants (c5032), expansion of education for immigrants (c5062), and positive statements about immigrants (c7053).

Together, these indicators capture the extent to which manifestos promote societal homogeneity, blame outgroups such as immigrants for criminality, and advocate restricting the rights of outgroups.

- (2) **Anti-institutionalism:** Populist discourse frequently presents constitutional constraints, intermediary institutions, experts, and political elites as obstacles to the authentic expression of the popular will. This discourse may also support the concentration of executive authority in a leader who claims to overcome such obstacles in the name of the people (Müller, 2016; Urbinati, 2019).

Operationalization: Anti-institutionalism is increasing in negative constitutionalism (c204), political authority (c305) and political corruption (c304), and decreasing in positive constitutionalism (c203).

Together, these indicators capture opposition to constitutional constraints and antagonism toward the political system, including portrayals of the system as rife with corruption and abuse, claims about the incompetence of other parties³⁸, and support for strong political authority.

- (3) **Sovereignism:** Populist sovereignism specifies what “the people” have supposedly lost: national authority, territorial control, borders, and protection from globalization (Basile & Mazzoleni, 2020). It advocates reclaiming the nation’s full control over its laws and economy. Recent scholarship highlights that populist rhetoric is unlikely to manifest without sovereignism (Verzichelli, 2020) and identifies protectionism and anti-globalization narratives as common features of contemporary populism (Gurieff & Papaioannou, 2022).

Operationalization: Sovereignism is increasing in anti-internationalism (c109) and protectionism (c406), and decreasing in pro-internationalism (c107) and anti-protectionism (c407).

Together with the other dimensions, these indicators may capture the salience of grievances concerning the national authority, territorial control, and economic protection supposedly lost by “the people” to the international system and globalization.

³⁸ This aspect is included in c305.

Principal Component Analysis (PCA)

Before estimating the PCA, I standardize these dimensions across the 121 coded Spanish parties so that differences in their original variances do not determine their contribution to the resulting components.

Table A2 presents the eigenvalues and factor loadings. The results support retaining the first principal component, which explains approximately 51 percent of the total variance. This component increases with anti-multiculturalism, opposition to equal rights for immigrants, other negative immigration framings, negative constitutionalism, support for strong political authority, references to political corruption, anti-internationalism, and protectionism. I interpret the first component as a measure of anti-institutional and exclusionary populist rhetoric.

Table A2: Principal Component Analysis Results.

Eigenvalues and Explained Variance				
Component	Eigenvalue	Difference	Proportion	Cumulative
Component 1	1.528	0.681	0.509	0.509
Component 2	0.847	0.222	0.282	0.792
Component 3	0.625		0.208	1
Principal Components				
Variable	Comp. 1	Comp. 2	Comp. 3	Unexplained
People-centric antagonism	0.606	-0.441	0.662	0
Anti-institutionalism	0.480	0.866	0.137	0
Sovereignism	0.634	-0.235	-0.737	0

Note: Variables are standardized.

The predicted first-component score is saved for each party, and it is weighted by the share of votes in each municipality and then standardized to create an indicator, as shown in Equation [3] of the main text.

2.1.3. Geographical distribution of dependent variables

Figures A5, A6 and A7 show the geographical distribution of the dependent variables by election year and grid cell in municipal elections.

Figure A5: Geographic distribution of RQ index post-2010.

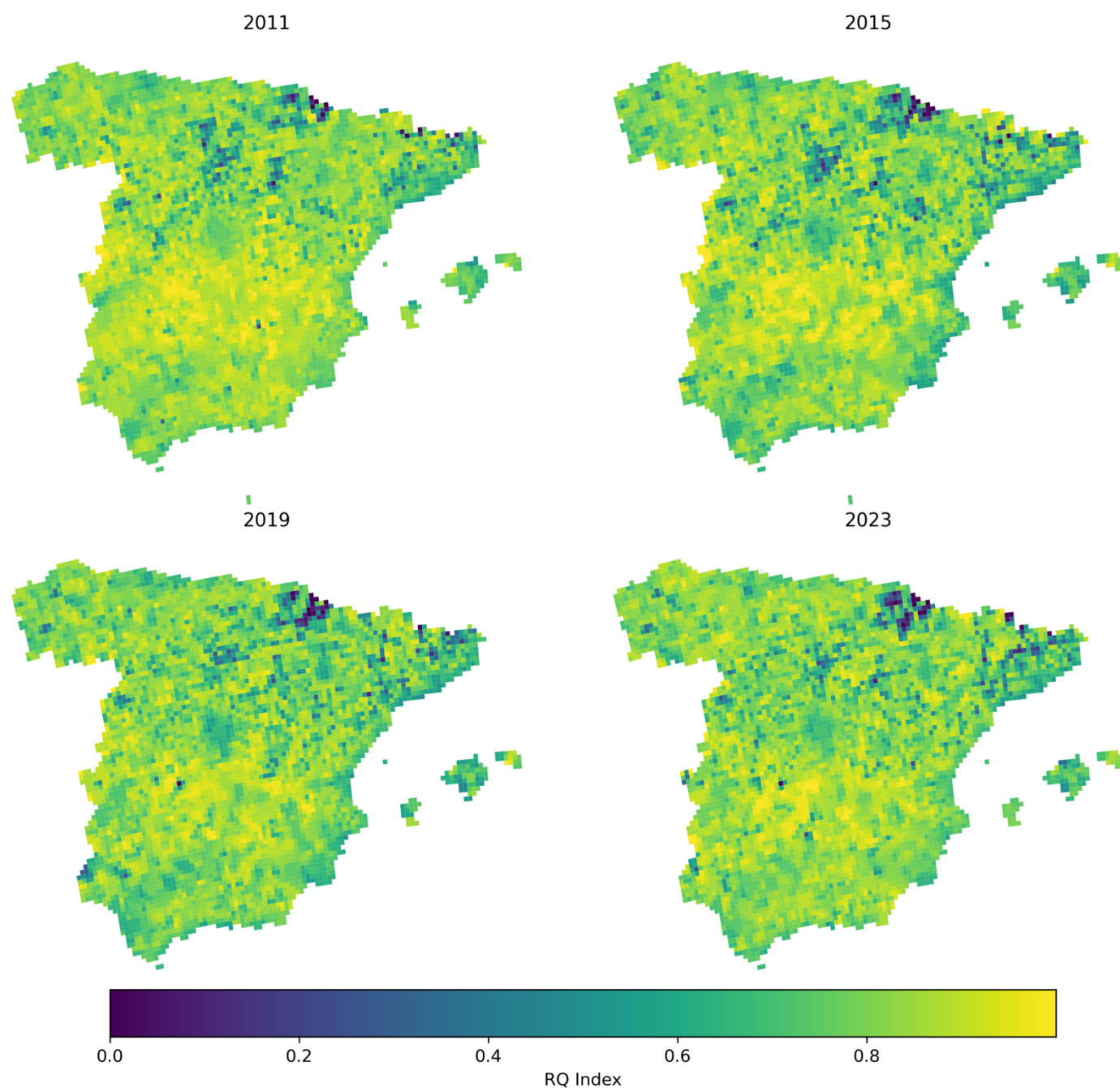


Figure A6: Geographic distribution of Dalton index post-2010.

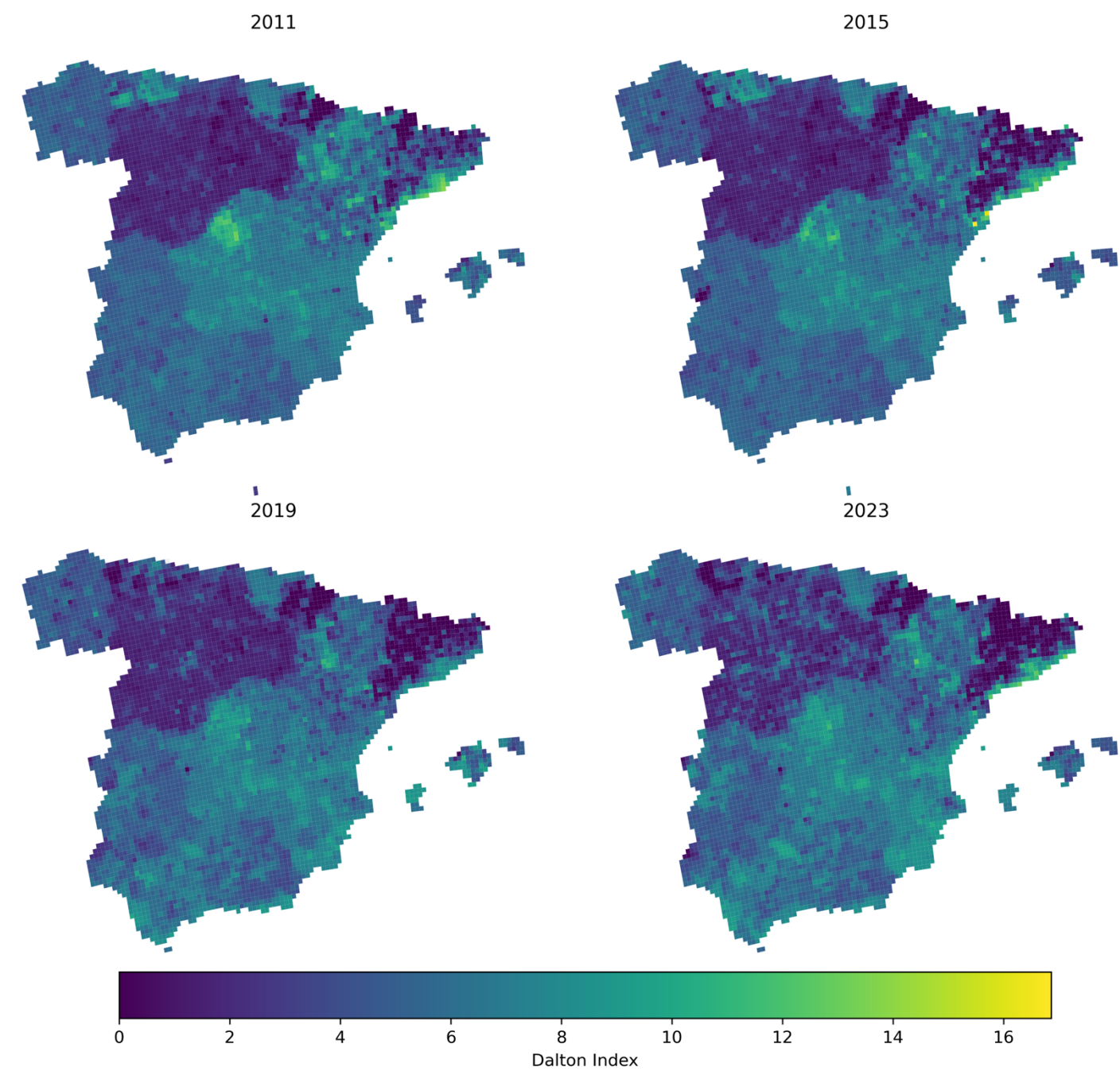
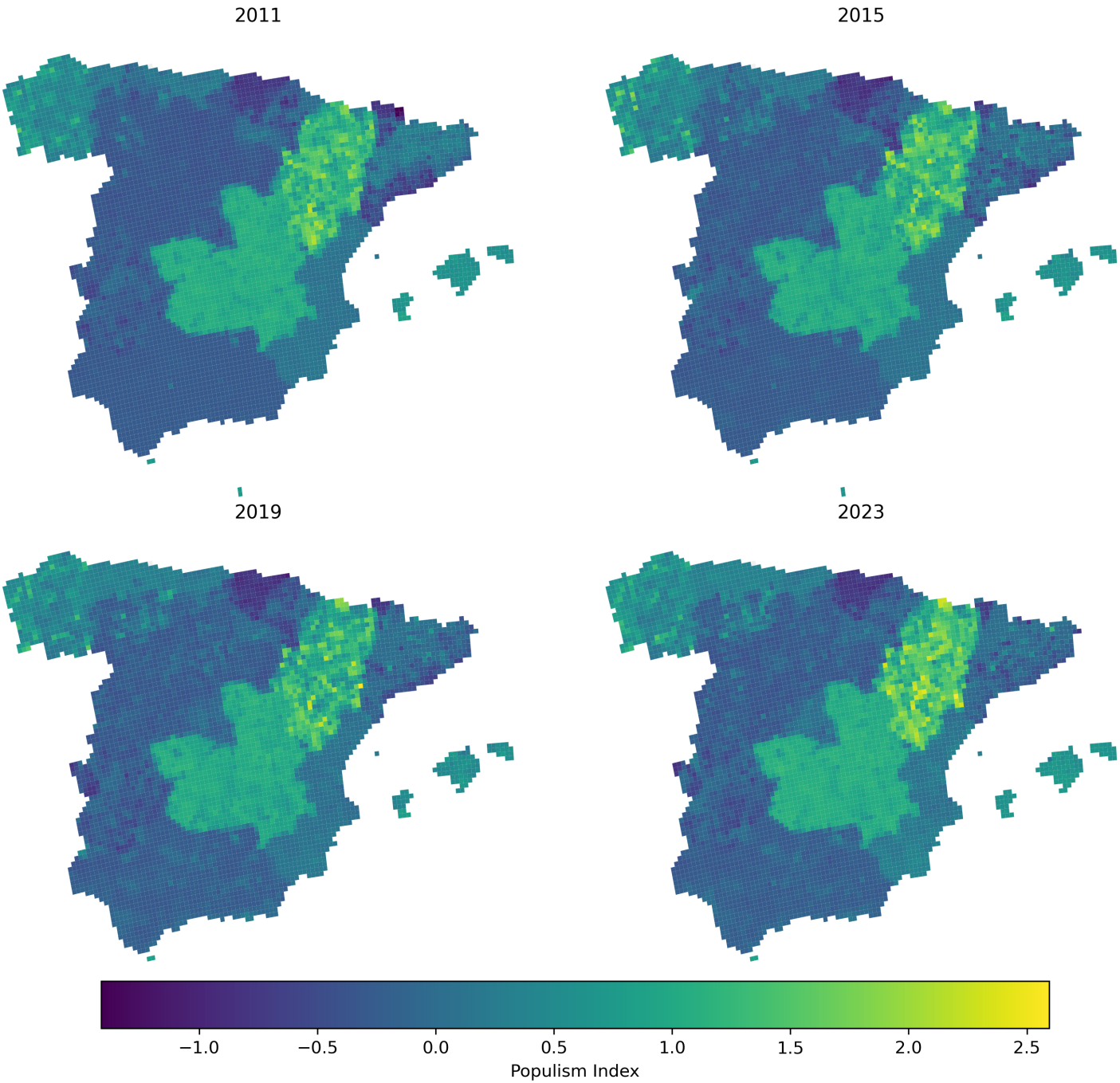


Figure A7: Geographic distribution of Populism index post-2010.



2.2. War exposure and instrumental variable

2.2.1. War exposure metrics

I attempt to compile all documented battles for each of the wars studied. To do so, I combine several sources that document these battles, including their locations and other indicators, such as casualties and outcomes, from the Wikidata entries associated with each war and its related subsections. These data were complemented and verified using the scholarly sources cited in Wikidata. The process yielded 290 battles for the Carlist Wars. The full list of consulted sources is recorded in a spreadsheet and is available upon request because of its length.

Georeferencing was conducted using a combination of coordinates from Wikidata, OpenStreetMap, and the ChatGPT API, followed by manual verification. Exposure is then defined as all grid cells located within a specified radius of each event, with the radius varying by specification.

2.2.2. Roads

Road indicators were constructed using the *Mapa general de caminos de España y de Portugal* (*General Map of Roads in Spain and Portugal*), which depicts the location of different types of roads in early nineteenth-century Spain and Portugal: mule roads (*Camino de Herradura*), royal roads (*Camino Real*), cart roads (*Camino Carretero*), and post roads (*Camino de Posta*). The map is hosted by the University of Utrecht. The different road types and their spatial distribution were then manually digitized. After dividing the data into grid cells, I calculate the length of each road segment located within each grid cell and use this as the road indicator. The digitized roads are shown in Figure A8.

Road-type descriptions

Mule roads (*Camino de Herradura*): Narrow roads suitable for pack animals (especially mules and horses) and pedestrians. Their width and construction generally prevented the passage of wheeled vehicles such as carts and wagons.

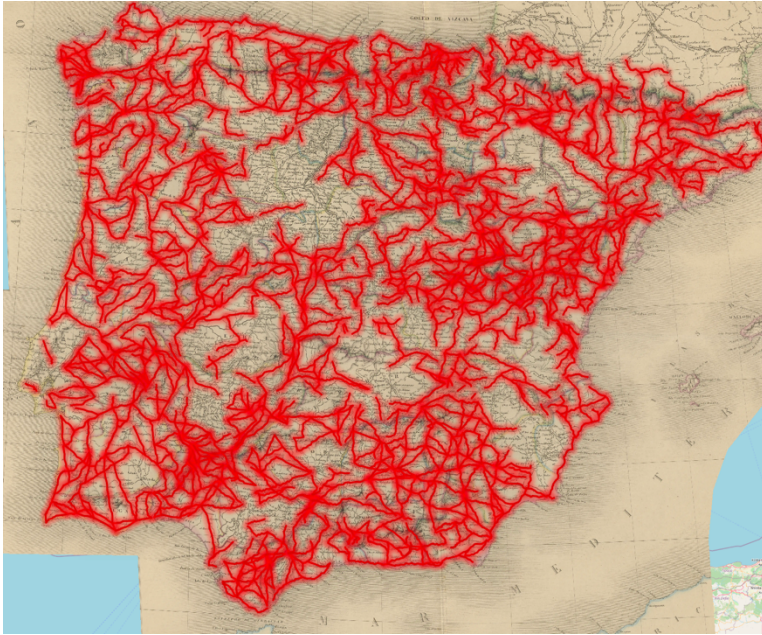
Post roads (*Camino de Posta*): Routes designated for postal services and official couriers. They were equipped with relay stations (*postas*) where travelers could change conveyances, facilitating the rapid transmission of mail and official correspondence.

Cart roads (*Camino Carretero*): Roads sufficiently wide and stable to permit the passage of wheeled vehicles, including carts and wagons.

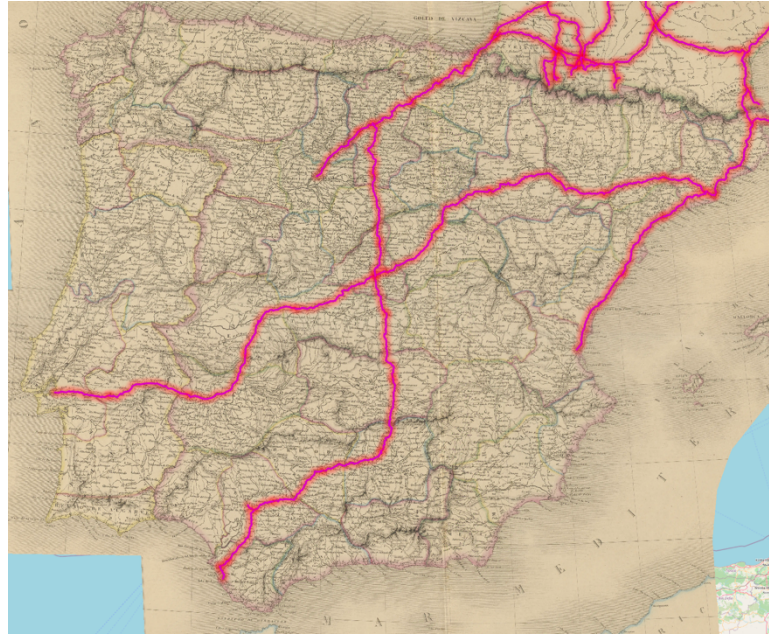
Royal roads (*Camino Real*): Principal roads of the kingdom operating under the authority and protection of the Crown. They connected major cities, ports, and frontier regions and played a central role in trade and administration.

Figure A8: Digitized roads by type in the Iberian Peninsula.

a) Mule roads.



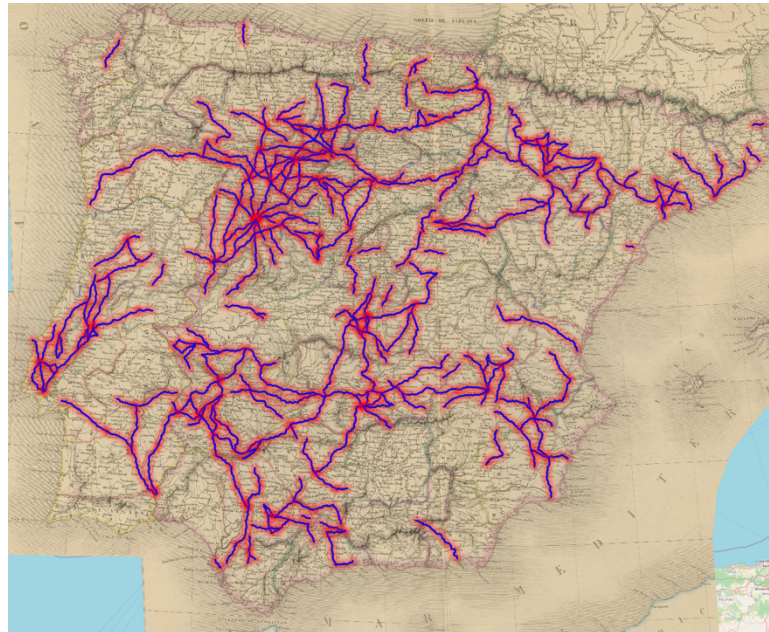
b) Post roads.



c) Royal roads.



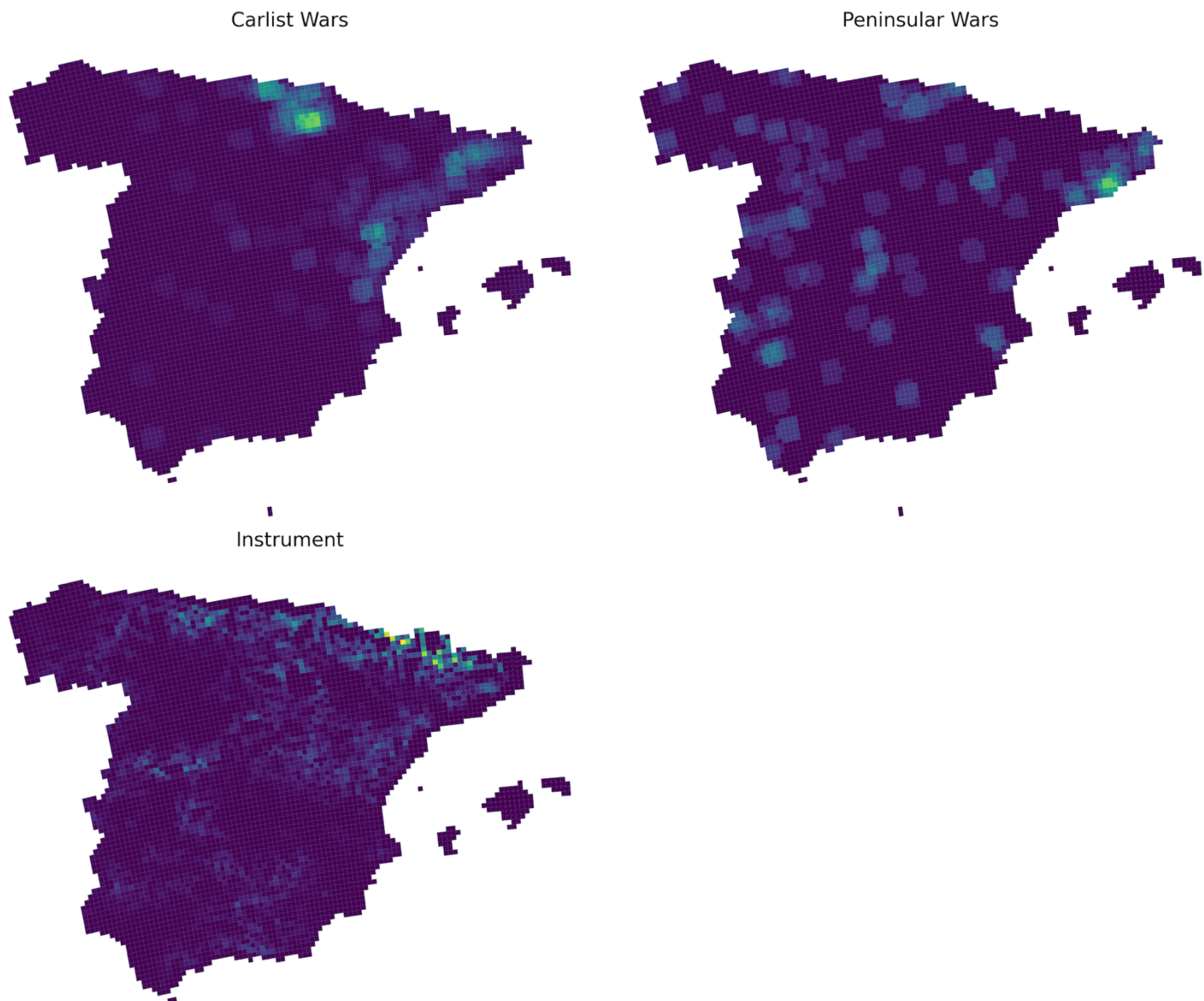
d) Cart roads.



2.2.3. Instrumental Variable

Figure A9 shows the geographic distribution of the instrumental variable across Spain, alongside the locations of Carlist Wars and Peninsular War battles. The distribution of Carlist Wars battles more closely mirrors that of the instrumental variable than does the distribution of Peninsular War battles. This pattern is consistent with the argument in the main text that logistical conditions varied across conflicts depending on the belligerents.

Figure A9: Distribution of Carlist Wars and Peninsular War battles and of the instrumental variable.



Note: All variables are standardized for comparability.

2.3. Construction of mediators

2.3.1. Cross-cutting cleavage indicators

I estimate the degree to which the cleavage structure is cross-cutting at the local level using two indicators, each appropriate for a different type of positional classification (categorical or continuous). For elections prior to 1936, I rely on categorical indicators constructed from party descriptions and coalition membership to infer party positions on the left-right and center-periphery dimensions. For elections after 1975, I use the RILE and center-periphery (CP) positions from the Regional Manifestos Project. To capture stable patterns in ideological dimensions that are less likely to reflect election-specific shocks, I average these measures over the pre-2000 period. I also use RILE and CP salience measures to address limitations of the cross-cutting indicator when positions are measured on continuous scales.

Ideological coding of parties before 1936: Parties were coded using two sources. First, I relied on Artola (1975), who provides an in-depth analysis of the history of political parties in Spain, including a synthesis of 112 political programs. Based on this source, I coded parties' positions on the left-right and center-periphery axes. In many cases, Artola explicitly describes parties as "left" or "right." In the remaining cases, I coded a party as "right" when it was described as conservative, monarchist, Catholic/confessional, anti-socialist, fascist, corporatist, anti-Marxist, or anti-liberal. I coded a party as "left" when it was described as anti-capitalist, collectivist, revolutionary, communist/anarchist, socialist, or when it used class-struggle language. I define the center as the residual category.

For the center-periphery dimension, I coded a party as "centralist" when it defended unitary sovereignty or centralized administration, or rejected regional autonomy. I coded a party as "regionalist" when it demanded greater administrative autonomy for the polity it represented.

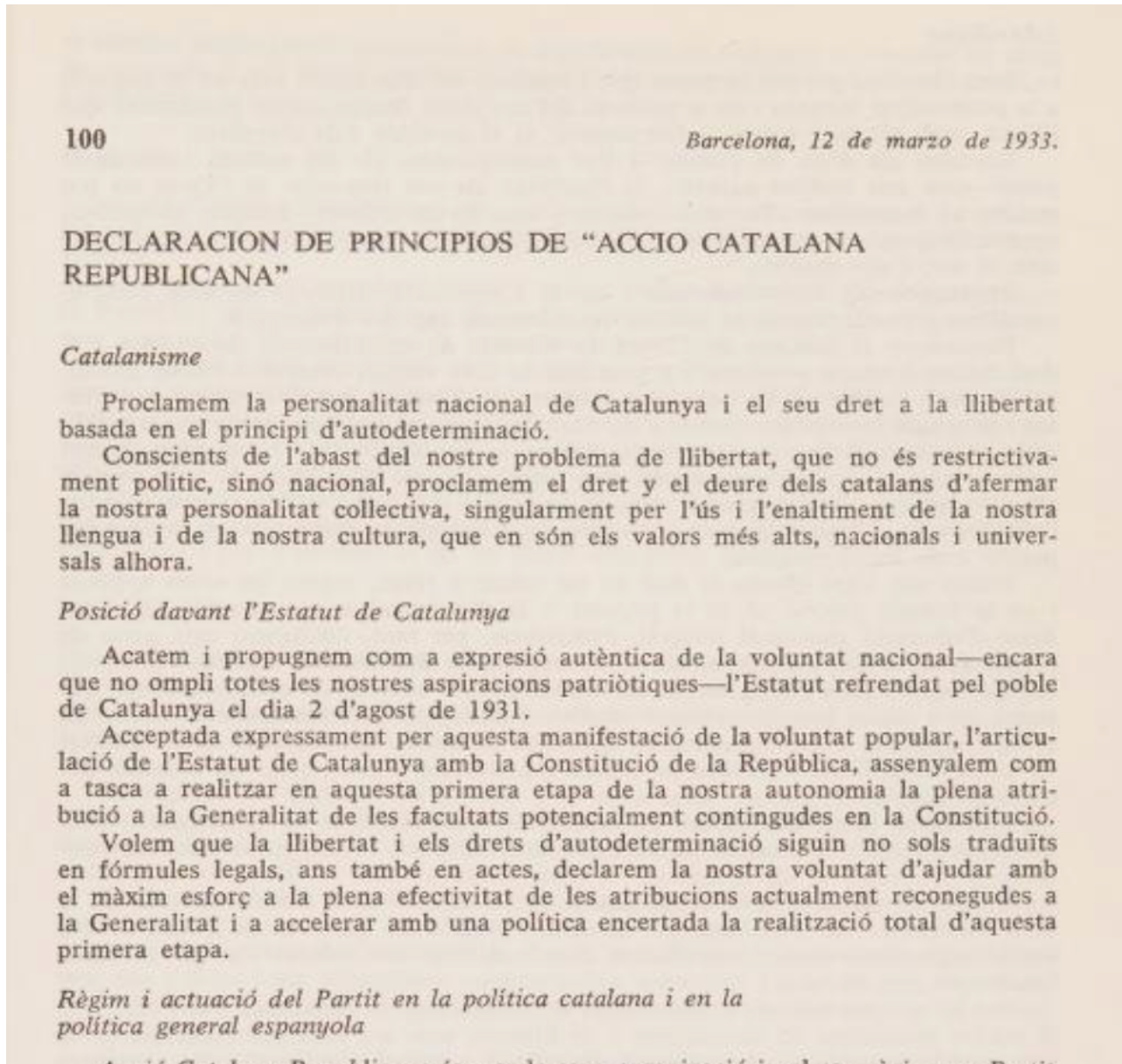
Second, I complemented this coding with information from *Historia Electoral*, including descriptions of the parties participating in each election and the coalitions through which they competed.

Examples of coding based on source and coalition

- a) Figure A10 shows the example of *Acció Catalana Republicana* in 1933. The text contains strong references to Catalan autonomy, placing the party on the periphery side of the center-periphery cleavage. The party can be traced back to *Partit Catalanista Republicà*; later, in 1936, it

appeared alongside left-wing parties such as *Esquerra Catalana* and *Front d'Esquerres*, placing it on the left of the left-right axis.

Figure A10: Example of description found in Artola (1975).



- b) Another illustrative example is *Derecha Regional Valenciana*, which translates as “Valencian Regional Right.” A party can be regional in name or organization without being located on the periphery side of the center-periphery axis. For instance, *Derecha Regional Valenciana* joined *Acción Popular* and other regional parties within CEDA. Because CEDA was a right-wing corporatist party that opposed greater regional autonomy, I code *Derecha Regional Valenciana* as occupying the center (centralist) position on the center-periphery axis.

Voting data

For elections up to 1931, I have data only for candidates and parties that were ultimately elected to Congress, rather than for all candidates who competed. These data come from the online Historical Record of Congresspeople on the official website of the Spanish Congress and were later complemented with information from *Historia Electoral*. Figure A11 shows an example of the Historical Record data, including the electoral constituency, district, political faction, election date, dates in office, number of votes obtained, number of eligible voters, and other information.

Figure A11: Example of data from the Historical Record of Congresspeople.

ACILLONA Y GARAY , JOSE DE

Datos generales

Título: MARQUES DE ACILLONA

Elecciones: 46. Elecciones 8.5.1910

Legislaturas: 1910-1911

Circunscripción: Vizcaya

Distrito: Marquina

Electores: 6888

Votantes: 4104

Votos obtenidos: 3102

Nº credencial: 346

Fecha de alta: 27/05/1910

Fecha de baja: 28/06/1910

Fracción política: Independiente

Profesión: Abogado

Using these data, I compute the percentage of elected congresspeople from each political faction in each constituency. After 1931, *Historia Electoral* also provides data on the votes obtained by all candidates in each constituency, which allows me to compute indicators of cleavage cross-cuttingness.

Formulas for the cross-cutting cleavage indicators

a) From 1931 to 1936

Because the variables identifying parties' positions on the left-right and center-periphery cleavage dimensions are coded as categorical variables, I compute the cross-cuttingness (XC) indicator proposed by Taylor and Rae (1969). The indicator in a specific location is defined as:

$$XC = \sum_{i=1}^m v_{i.}^2 + \sum_{j=1}^n v_{.j}^2 - 2 \sum_{i,j} v_{ij}^2$$

where i indexes positions on the left-right dimension—left, center, and right—and j indexes positions on the center-periphery dimension—centralist, peripheral, and neither. The term $v_{i.}$ denotes the vote share of all parties in category i on the left-right dimension, while $v_{.j}$ denotes the vote share of all parties in category j on the center-periphery dimension. Finally, v_{ij} denotes the vote share of parties that belong simultaneously to category i on the left-right dimension and category j on the center-periphery dimension.

Substantively, the XC indicator captures the probability that two randomly selected voters share the same position on one cleavage dimension but differ on the other. Thus, higher values of XC indicate greater cross-cuttingness between the two cleavage dimensions.

b) From 1975 onwards

For elections after 1975, I use the RILE and CP positions and saliency indicators from the Regional Manifestos Project. Because these variables are continuous, I use the complement of the issue alignment indicator proposed by Baldassarri and Gelman (2008). The latter corresponds to the correlation between one ideological dimension and the other. Higher values of the complement indicate greater cross-cuttingness because they mean that position on one cleavage dimension is less predictive of position on the other.

The problem with this indicator for the purposes of my study is that the correlation may be high even when these dimensions are not important, which overstates the degree of cleavage cross-cuttingness. This does not happen with elections before 1936 because the ideological components are constructed based on coalitions or scholarly accounts; if a dimension is not particularly relevant, it will not appear in the scholarly treatment of the party, or the coalitions parties join.

To address this problem for post-1975 elections, I draw on Ansolabehere and Puy’s (2018) distinction between the electoral salience of an issue and the configuration of party positions on that issue. In their framework, issue salience weights the contribution of each ideological dimension to voters’ utility, so the electoral consequences of party positions depend jointly on issue salience and party differentiation³⁹. Motivated by this framework, I scale positional cross-cuttingness by the joint salience of the two cleavage dimensions:

$$CC = \left(1 - \left| \frac{\sum_j v_j (RILE_j - \overline{RILE})(CP_j - \overline{CP})}{\sqrt{\sum_j v_j (RILE_j - \overline{RILE})^2} \sqrt{\sum_j v_j (CP_j - \overline{CP})^2}} \right| \right) \cdot \sqrt{\sum_j v_j SRILE_j \cdot \sum_j v_j SCP_j}$$

Here, j indexes parties. $RILE_j$ denotes party j ’s RILE position, and CP_j denotes its center-periphery position. $SRILE_j$ and SCP_j denote the salience of the $RILE$ and CP dimensions in party j ’s program, respectively. Finally, v_j denotes party j ’s vote share in that location. $\overline{RILE}$ and $\overline{CP}$ denote the vote-weighted average RILE and CP positions in that location, such that $\overline{RILE} = \sum_j v_j RILE_j$, and $\overline{CP} = \sum_j v_j CP_j$.

Thus, the indicator is highest when, on average, parties’ left-right positions do not strongly predict their center-periphery positions and both dimensions are important for voting decisions.

2.3.2. Regional parties

Another mechanism I examine is whether affected areas were more likely to create regional parties and support them electorally. This would indicate a greater propensity for political articulation, regardless of whether such articulation was centralist or peripheral. By comparing the creation and subsequent electoral support of regional parties across areas with different levels of exposure, I assess whether affected areas were more likely to participate politically through party formation and support.

For elections before 1936, I compile a set of 277 parties and movements from a combination of sources, including Artola (1975), the Historical Record of Congresspeople combined with Historia Electoral, and Wikidata, following the verification procedure described for war exposure. Parties are classified as regional or national depending on their geographic reach. The full list of parties is recorded and available upon request. This compilation shows that all Spanish regions created regional parties at some point

³⁹ See Ansolabehere and Puy (2018) for further details on this approach.

before 1936, although not all succeeded in winning office. I exploit this variation and estimate the effect on the number of elected congresspeople from regional parties as a mediator.

For elections after 1975, I define a regional list as one that participates in elections in at most four autonomous communities. This cutoff reflects a discrete break in the distribution: raising the threshold to five or six autonomous communities does not add any additional lists.

The remaining mediators—spouse geographic similarity, the Dalton index of the birth-baptism interval, and commercial organizations—are measured shortly after the wars ended. Since these variables are also used as controls when measured before the war, they are discussed in the next section.

2.4. Construction of control variables

2.4.1. HYDE data

The History Database of the Global Environment (HYDE) is a spatially explicit database of historical population and land use maintained by the Netherlands Environmental Assessment Agency. I use HYDE 3.5, which has a spatial resolution of 5 arc-minutes, to obtain measures of urban built-up area and irrigation in 1830, immediately before the beginning of the Carlist Wars. I also use HYDE population estimates for 1780 and 1830 to construct local population growth rates.

Because HYDE population figures are reconstructed estimates, I anchor population levels to the observed 1787 census. I then estimate population in each location in 1830 by applying the annualized population growth rate implied by HYDE between 1780 and 1830 to the population observed in the 1787 census.

2.4.2. Market potential indicator

I include a measure of pre-war market potential to compare locations that were similarly connected to large commercial hubs. To construct this metric, I follow Harris (1954), who proposes an index equivalent to the sum of market size divided by the cost of accessing each market. In this setting, market potential is defined as:

$$MP_i = \sum_{j=1}^N \frac{Pop_j}{C_{ij}}$$

Here, i is the location for which market potential is computed, j corresponds to a commercial hub, Pop_j corresponds to its population, C_{ij} is the cost of moving from location i to city j and N is the number of commercial hubs used in the analysis. I restrict the set of commercial hubs to locations with populations above 5,000, rather than using all locations in the sample. This restriction makes the measure capture access to commercial centers more clearly, rather than local population density. The resulting measure uses 458 commercial hubs. Table A3 shows the ten largest hubs.

Table A3: Estimated ten largest commercial hubs in Spain by 1830.

Pos.	Hub	Pos.	Hub
1	Madrid	6	Granada
2	Valencia	7	Pamplona
3	Barcelona	8	Murcia
4	Sevilla	9	Málaga
5	Cádiz	10	Zaragoza

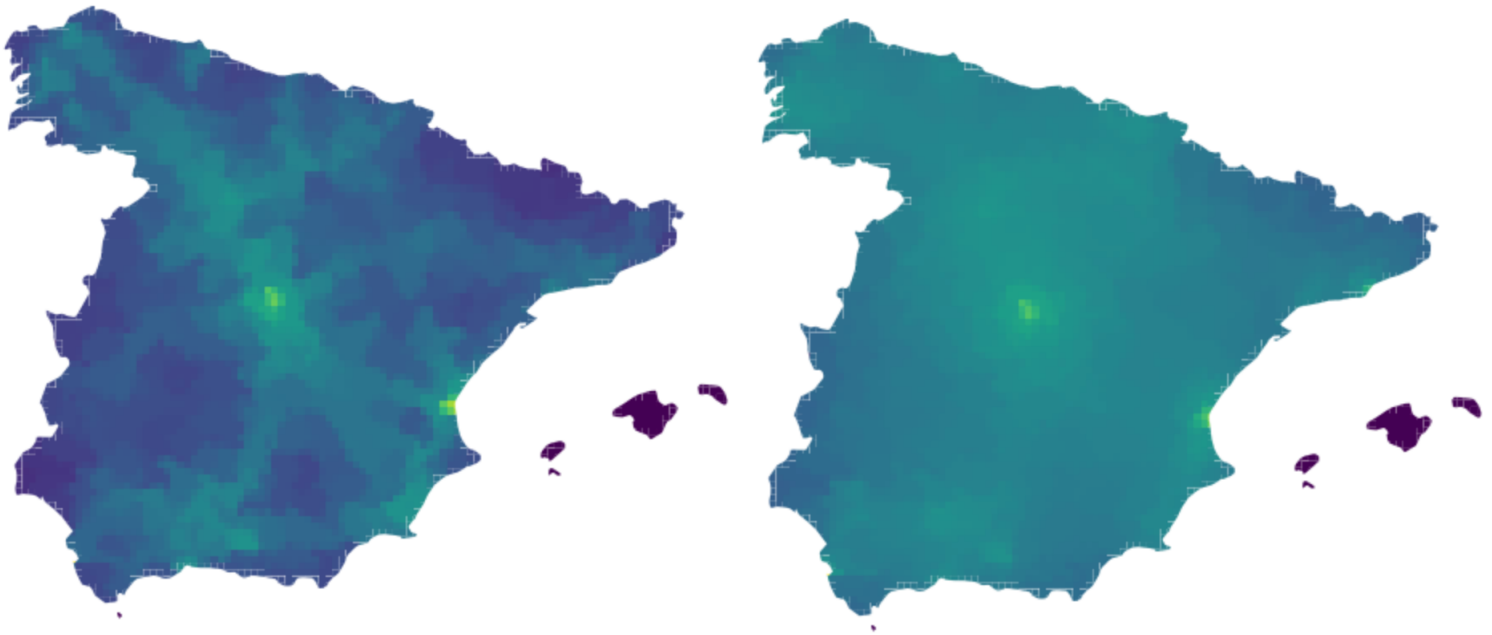
Notes: “Pos.” refers to the relative position, with 1 being the largest commercial hub.

I compute transportation costs in two ways. First, I use the average travel time implied by the existing road network. I proxy travel speeds using evidence from the United Kingdom and Peru for roads comparable to each road type in my setting (Zegarra, 2011; Bogart et al., 2019; Rosevear et al., 2019; García-López et al., 2023). Second, I use Euclidean distance as the transportation cost, following Harris (1954). Figure A12 shows both versions of the market-potential measure.

Figure A12: Market potential by 1830.

a) Cost as time of transportation

b) Cost as Euclidean distance



2.4.3. Commercial organizations

I use commercial organizations both as a control variable, when measured before the war, and as a mediator, when measured after the war.

To construct this indicator, I digitize the *Censo Electoral de Asociaciones Profesionales y Relación de las Instituciones no Profesionales de Ahorro, Cooperación y Previsión*. This source lists professional associations, savings and pension institutions, and cooperation institutions in Spain, including their creation dates and geographical locations. Figure A13 provides an example from the original source.

Figure A13: Example of social organizations, creation dates and locations.

Córdoba.

- 51. Cámara oficial agrícola. 25 agosto 1906. — Córdoba.**
- 52. Cámara agrícola oficial. 17 febrero 1907. — Fuenteovejuna.**
- 53. Cámara agrícola. 10 marzo 1906. — Lucena.**
- 54. Cámara agrícola oficial. 23 julio 1908. — Viso.**

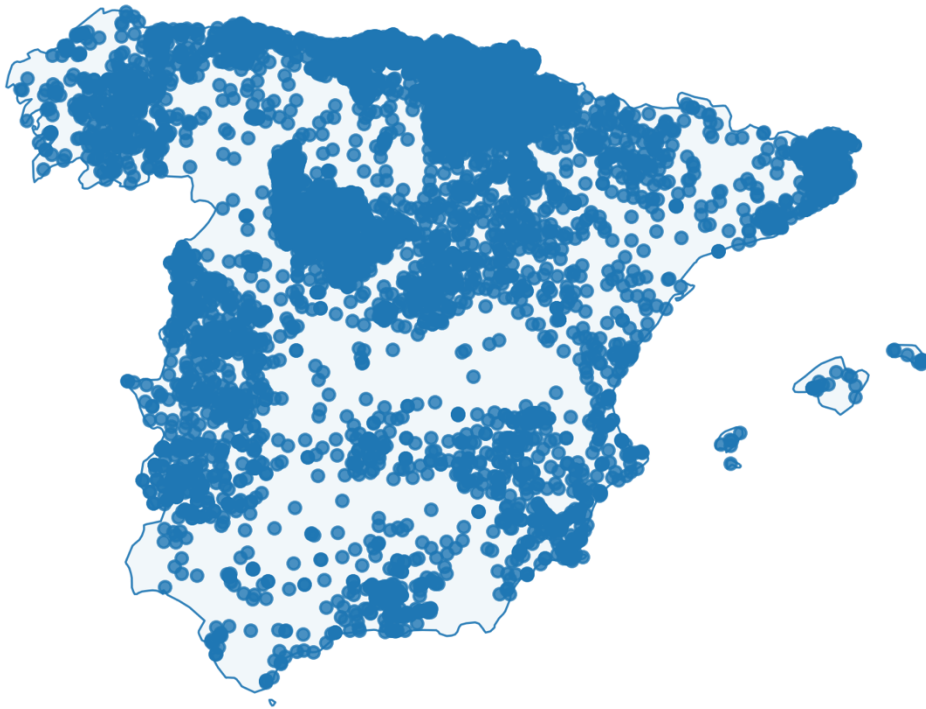
This digitization yields more than 14,000 organizations created by 1916. I then construct an indicator equal to the number of organizations created up to a given year, divided by the corresponding

population. For example, the number of organizations by 1900 is the number of organizations with a recorded creation date before 1900.

2.4.4. Indicators from baptism records

For the case of Spain, the analysis is conducted on a sample of 601,746 baptism records. These contain the location of the baptism, the full name of the baptized, the dates of baptism and birth, as well as, in some cases, the names of the parents. These are used as controls when measured before the wars, and as mediators when measured after. Figure A14 shows the locations of the baptism places recorded.

Figure A14: Locations of georeferenced baptism places.



When a grid cell had more than one baptism place, the weighted average of each of the created indicators was computed using as weights the number of baptisms in each location. For grid cells without baptism places recorded, the indicator for the closest baptism site was used. To account for this, I also compute the distance from the grid centroid to the closest baptism place for all grid cells. Next, I discuss the indicators that were created using this data source.

Birth-baptism time interval: For each baptism record, I compute the number of days elapsed between the child’s date of birth and date of baptism. I then aggregate this variable at the baptism-site level in two ways. First, I compute the average delay between birth and baptism. Second, I compute a Dalton-style dispersion measure, equivalent to the standard deviation of baptism delays within the baptism site.

The average delay captures the typical speed with which families baptized children, while the dispersion measure captures heterogeneity in this practice. I interpret these variables as proxies for religious adherence and religious heterogeneity, respectively. The interpretation relies on Catholic doctrine, which prescribed early baptism because of the spiritual consequences attached to dying unbaptized⁴⁰. Families placing greater weight on this belief should have stronger incentives to baptize children soon after birth. Variation in the timing of baptism can therefore reveal differences in religious urgency, although it may also reflect access costs. For this reason, the empirical specifications also control for distance to the nearest baptism site.

This interpretation is consistent with historical evidence showing that baptism delays responded to changes in infant mortality (Marco-Gracia, 2022). When early death became less likely, families had weaker incentives to baptize immediately, suggesting that baptism timing reflected perceived religious and mortality-related urgency rather than only administrative convention.

Spouse geographic similarity indicator: I also construct an indicator that proxies the importance of preserving geographically rooted lineage or identity using Spanish naming conventions, under which children generally inherit surnames from both parents. The measure captures whether the paternal and maternal surnames of baptized children were historically concentrated in similar geographic areas.

The construction proceeds in three steps. First, I use baptism records before the war to estimate the historical spatial distribution of each surname across baptism sites. For surname s , I define:

$$P(c|s) = \frac{f_{sc}}{\sum_c f_{sc}}$$

⁴⁰ The Council of Trent, for instance, states that “*Since infant children have no other means of salvation except Baptism, we may easily understand how grievously those persons sin who permit them without the grace of the Sacrament longer than necessity may require, particularly at an age so tender as to be exposed to the numberless dangers of death*” (Council of Trent, 1566, p. 129).

where f_{sc} is the number of occurrences of surname s in baptism site c during the baseline period. This produces, for each surname, a vector describing its historical geographic distribution.

Second, for each baptism record, I retrieve the vectors corresponding to the father's and mother's first surnames, denoted $\mathbf{p}_f$ and $\mathbf{p}_m$. I then compute the cosine similarity between the two vectors:

$$\text{Cosine Similarity} = \frac{\mathbf{p}_f \cdot \mathbf{p}_m}{\|\mathbf{p}_f\| \|\mathbf{p}_m\|}$$

This score ranges from 0 to 1. Higher values indicate that the paternal and maternal surnames had similar historical geographic distributions, while lower values indicate that they were historically concentrated in different places.

Third, I average this record-level similarity score at the baptism-site level. The resulting indicator measures the extent to which families in a location were formed by parents whose surnames had similar geographic origins. Higher values are interpreted as greater endogamy or local origin concordance; lower values indicate greater exogamy, migration, or more open marriage networks.

This approach has two advantages. First, it relies on the empirical distribution of surnames rather than on ex ante classifications of surnames into regions or ethnic groups. Second, because the surname distributions are fixed using the pre-war baseline, later values of the indicator capture changes in the pairing of surnames rather than changes in the definition of surname origins themselves.

Nevertheless, the constructed indicators are arguably more useful for roughly capturing cross-sectional differences across the country over long time windows—for example, as control variables—than for detailed analyses of religious or family-forming behavior over time, since they are based on a sample of available records rather than the full population.

3. Annex references

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